The Huang–Li reduction of Goldbach to Elliott–Halberstam does not decompose, with an unconditional bound for a Möbius-weighted correlation sum in fixed residue classes

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Abstract

Huang and Li [2] show that the binary Goldbach conjecture for large even N follows from EH together with a Möbius-twisted variant EH_μ whose levels sum to more than 1; by Bombieri–Vinogradov their Corollary 1 reduces this to $EH_\mu(N^{\theta'})$ alone for a single $\theta' > 1/2$, consumed at two places in their §3, the functionals they call $E_4(\alpha)$ and $E_3(\alpha)$. We show that this reduction does not decompose: of the two consumptions in that section there is no half whose discharge would constitute partial progress. One of the two is unnecessary — the Möbius-weighted correlation sum underlying E_4 is $\ll_A N(\log N)^{-A}$ for every $A > 0$ unconditionally and uniformly in the truncation point (Theorem 1), so that section's EH_μ demand collapses to the single scalar $E_3(\alpha)$ (Corollary 2) — and the one that survives is not an equidistribution statement one might hope to establish separately: by an unconditional identity (Theorem 3) it is Huang–Li's own equation (22). It therefore already yields binary Goldbach for large even N, and yields the asymptotic $\tilde{r}(N) \sim \mathfrak{S}(N)N$ exactly when $\sum_{n<N} \Lambda(n)\mu(N-n) = o(N)$. After the half that can be discharged has been discharged, what is left is the conclusion (Note 5). The root cause is $\mu * \log = \Lambda$: the weight $\log k$ carries the Goldbach content, so every divisor switch must hand it back.

We then locate the true strength of the surviving demand: Goldbach needs only a one-sided bound on E_3 (Proposition 7), at a threshold of order N over the even N measured, and hence, off the thin family where that threshold falls by a factor $\log N$, only a constant-factor bound where EH_μ is consumed with a saving of $(\log N)^A$ for every A (Proposition 8). We also treat a defect in [2] that is not ours to report: equation (18) drops an n-dependent constraint present in the definition of $S_2(\alpha)$, which S. Zheleznov reported to the authors independently and earlier, and which Moya [14] both exhibits by an exact finite counterexample and repairs under [2]'s own hypothesis by keeping the moving endpoint. What is done here is the other side of it: the missing term is *bounded* against the published form and shown harmless under the hypotheses already assumed, which is what carries Theorem 1's bound over to the corrected formulation the authors have since adopted (Proposition 25).

We state plainly what is not claimed. Theorem 1 removes only the part of the demand that carries no Goldbach content, and the net progress toward the Goldbach conjecture is zero. We also state plainly what is standard: the ingredients of Theorem 1 are classical and its mechanism — a divisor switch that moves the work onto a short variable, followed by Bombieri–Vinogradov — is ordinary practice. What is offered is the application, the identity it makes visible, and the corrections both turn out to require. The finite rearrangements the argument runs on are formalised in Lean 4 against Mathlib, over arbitrary functions, so that what is checked is that they use no arithmetic property of Λ or μ.

1 Introduction

1.1 The Huang–Li reduction

Let N be a large even integer, Λ the von Mangoldt function, μ the Möbius function, and

$$\tilde{r}(N) \;=\; \sum_{n<N} \Lambda(n)\Lambda(N-n), \qquad \mathfrak{S}(N) \;=\; 2\prod_{p>2}\Big(1-\tfrac{1}{(p-1)^2}\Big)\prod_{\substack{p\mid N\\ p>2}}\Big(1+\tfrac{1}{p-2}\Big).$$

Write $EH_\mu(Q)$ for the assertion that for every $A>0$

$$\sum_{q\le Q}\max_{y<N}\max_{(a,q)=1}\Big|\sum_{\substack{n\le y\\ n\equiv a\ (q)}}\Lambda(n)\mu(N-n)-\frac{1}{\varphi(q)}\sum_{n\le y}\Lambda(n)\mu(N-n)\Big| \ll_A \frac{N}{(\log N)^A}. \tag{1}$$

Huang and Li [2], continuing Pan [11], prove that $EH(N^\theta(\log N)^{2A+8})$ together with $EH_\mu(N^{1-\theta})$ implies

$$\tilde{r}(N) \;\ge\; \mathfrak{S}(N)\big(1-\mathfrak{A}(N)\big)N + O\Big(\frac{N}{(\log N)^A}\Big), \tag{2}$$

and deduce (their Corollary 1) that, in view of Bombieri–Vinogradov, binary Goldbach for all sufficiently large even integers follows from $EH_\mu(N^{\theta'})$ alone for any single $\theta'>1/2$. Here

$$\mathfrak{A}(N) \;=\; \prod_{\substack{q\nmid N\\ q>2}}\Big(1-\frac{1}{q(q-1)}\Big) \tag{3}$$

is their equation (7); N is even, so $q>2$ is implied by $q\nmid N$ and is written here only to match their display. We write $\mathfrak{A}$ rather than their A because A is already the free exponent in (1), and one symbol must not carry two meanings.

This is the sharpest conditional frame we are aware of: one sentence stands between classical technology and Goldbach. This paper asks what that sentence costs on the side where it is consumed.

1.2 Where EH_μ is consumed

The hypothesis (1) enters their §3, where the reduction is carried out, at two places. That those two are all of them is a statement about that section and is read off the argument there; no enumeration of [2] outside §3 is offered here, and the statements below that quantify over the consumptions — the collapse of Corollary 2 and the non-decomposition of Note 5 — carry that scope with them. With α the Vaughan-type parameter of their §3 and $K=(N-1)/\alpha$, these are (their notation; note that the residue class is the *fixed* class $a\equiv N$):

$$E_3(\alpha) = \sum_{\substack{k<K\\ (k,N)=1}}\mu(k)\log k\Big[\sum_{\substack{n<N\\ n\equiv N\ (k)}}\Lambda(n)\mu(N-n)-\frac{1}{\varphi(k)}\sum_{n<N}\Lambda(n)\mu(N-n)\Big], \tag{4}$$

$$E_4(\alpha) = \sum_{\substack{k<K\\ (k,N)=1}}\mu(k)\Big[\sum_{\substack{n<N\\ n\equiv N\ (k)}}\Lambda(n)\mu(N-n)\log(N-n)-\frac{1}{\varphi(k)}\sum_{n<N}\Lambda(n)\mu(N-n)\log(N-n)\Big]. \tag{5}$$

In both cases Huang–Li discard the signs $\mu(k)$ by the triangle inequality on the first line and then appeal to (1) (their Lemma 5 for E_4). The two functionals differ in exactly one respect:

E_3 carries the weight $w_k = \log k$ and E_4 carries $w_k = 1$ (the factor $\log(N-n)$ inside E_4 is k-independent and is removed by partial summation). This paper shows that this one difference decides everything.

Two parametrisations of α occur below and are named throughout. The *Corollary-1 regime* is $\alpha = N^{1-\theta'}$ with $\theta' \in (1/2, 1)$, so that $K = (N-1)/\alpha \asymp N^{\theta'}$ and the cofactor α that the completion leaves is below $N^{1/2}$; it is the parametrisation [2]'s Corollary 1 uses, and it is where the statements below are unconditional. The *Theorem-1 regime* is $\alpha \asymp N^{\theta}$, with θ the exponent of the level hypothesis their Theorem 1 assumes; there the same statements carry that hypothesis. We write δ for the quantity fixed by $N^{1-\theta'} = N^{1/2-\delta}$, that is $\delta = \theta' - \frac{1}{2} > 0$, so that the cofactor is $N^{1/2-\delta}$; and C_2 for the twin-prime constant $\prod_{p>2}\bigl(1-(p-1)^{-2}\bigr)$, so that $\mathfrak{S}(N) = 2C_2 \prod_{p\mid N,\, p>2}\bigl(1+(p-2)^{-1}\bigr)$.

1.3 Results

For $(k, N) = 1$ and $1 \le t < N$ put

$$E_\mu(t;k) := \sum_{\substack{n\le t\\ n\equiv N\ (k)}} \Lambda(n)\mu(N-n) - \frac{1}{\varphi(k)} \sum_{n\le t} \Lambda(n)\mu(N-n), \qquad C(t) := \sum_{n\le t} \Lambda(n)\mu(N-n), \quad (6)$$

and, for a weight w,

$$T_w(t) \;:=\; \sum_{\substack{k<K\\ (k,N)=1}} \mu(k)\, w(k)\, E_\mu(t;k).$$

Thus E_4 is obtained from T_1 and E_3 from $T_{\log}$. We write $E_\mu(N;k) := E_\mu(N-1;k)$, so that the argument N can name the whole range of (4) without calling for $\mu(0)$.

Theorem 1. *Fix $\theta' \in (1/2, 1)$ and put $K = N^{\theta'}$. Then for every $A > 0$,*

$$\sup_{1\le t<N} \bigl|T_1(t)\bigr| \;=\; \sup_{1\le t<N} \Bigl|\sum_{\substack{k<K\\ (k,N)=1}} \mu(k)\, E_\mu(t;k)\Bigr| \;\ll_{A,\theta'}\; \frac{N}{(\log N)^A}.$$

Moreover every ingredient of the proof except Bombieri–Vinogradov contributes only $O\Bigl(Ne^{-c\sqrt{\log N}}\Bigr)$: the power of $\log$ in the statement is imposed by Bombieri–Vinogradov alone.

Theorem 1 is a statement about a *signed* sum in a *fixed* residue class. It is not an instance of, and does not imply, $EH_\mu(N^{\theta'})$, which asks for absolute values and a maximum over residue classes. The mechanism is described in Section 2: after divisor switching one *completes* the divisor sum, and on the complementary range $k \ge K$ the Möbius factor on the long variable squares itself away while the surviving Möbius sits on a variable of size $\le N^{1/2-\delta}$. The completion is not a formality — without it the cofactor is not short and the configuration is the one with no known machine.

Corollary 2. *In the Corollary-1 regime, $E_4(\alpha) \ll_A N(\log N)^{-A}$ for every $A > 0$, unconditionally. Consequently Huang–Li's estimate $S_4(\alpha) = O(N(\log N)^{-A})$ requires no hypothesis, their Lemma 5 is not needed, and the EH_μ demand of their §3 collapses to the single scalar $E_3(\alpha)$ — together with the term Δ of Section 7 where the published form of (18) is used, which is likewise unconditional there, and which does not arise on the corrected form.*

Theorem 3. *In the Corollary-1 regime one has the unconditional identity*

$$E_3(\alpha) \;=\; \sum_{n<N} \Lambda(n)\Lambda(N-n) \;-\; \mathfrak{S}(N)\Bigl(N - \sum_{n<N} \Lambda(n)\mu(N-n)\Bigr) \;+\; O_A\Bigl(\frac{N}{(\log N)^A}\Bigr).$$

In particular the bound $E_3(\alpha) \ll_A N(\log N)^{-A}$ — the form of EH_μ that the Huang–Li argument consumes, though not the weakest that suffices; see Proposition 7 — is equivalent *to Huang–Li's equation (22). It therefore yields binary Goldbach for large even N through their Theorem 1; and, granted it, the asymptotic $\tilde{r}(N) \sim \mathfrak{S}(N)N$ holds if and only if $C(N) = o(N)$.*

Note 4 (three statements, not two)*.* The identity says that the displayed difference *is* $E_3(\alpha)$, so the hypothesis $E_3 \ll_A N(\log N)^{-A}$ and Huang–Li's (22) are the same assertion. What [2] record immediately after their (22) is a different equivalence, between $\tilde{r}(N) \sim \mathfrak{S}(N)N$ and $C(N) = o(N)$. Chaining the two, $E_3 \ll_A N(\log N)^{-A}$ delivers binary Goldbach — the bound $|C(N)| \le \mathfrak{A}(N)N(1 + o(1))$ of [2, §3.3], classically Mirsky's theorem [4], supplies the rest, with $\mathfrak{A}(N) < 1$; their Theorem 1 is the lower bound (2), which runs the other way and is a consequence of that bound rather than a source of it — but it does not on its own deliver the asymptotic, which needs $C(N) = o(N)$ in addition. The three statements must be kept apart.

Theorem 3 says that the demand side is closed at the level of *identities* rather than of *estimates*: no choice of θ', of truncation, or of smoothing can evade it, because the root cause is the identity $\mu * \log = \Lambda$ from which Huang–Li start. The weight $\log k$ is the carrier of the Goldbach content, so every divisor switch must hand it back.

Note 5 (what the collapse is, and what it is not)*.* It is not a weakening of the hypothesis. Huang–Li's Corollary 1 assumes a single $EH_\mu(N^{1-\theta})$ and spends it at two places; Corollary 2 says one of the two did not need spending, and the level demanded is unchanged, because E_3 still demands it.

What the collapse buys is a statement about the *shape* of the reduction rather than its strength, and it is visible only together with Theorem 3. After Theorem 1 there is, in their §3, exactly one place where the hypothesis is consumed, and by Theorem 3 that one place is not an equidistribution statement one might hope to establish separately: it is, by an unconditional identity, Huang–Li's own equation (22). **So the reduction does not decompose.** There is no half of E_μ that can be discharged to make partial progress — after the half that can be discharged has been, what is left is the conclusion. The two statements are meant to be read together, and a reader deciding whether to spend effort on this route is the reader they are for.

Note 6 (what is new in Theorem 3, and what is not)*.* The chain behind it is not ours. Murty and Vatwani [5] run it for the shift $n \mapsto n+h$: they decompose $\Lambda = \mu * \log$, split at y, carry the factor $\mu^2(n+h)$, and reduce to sums they call S_3 and S_4 — S_3 being exactly the log-weighted Möbius correlation sum in the fixed class, that is E_3. Assuming EH_Λ and their shifted-Möbius EH_{μ_h} they obtain $\sum_{n \le x} \Lambda(n)\Lambda(n+h) \sim (\mathfrak{S}(h)+o(1))\big(x-\sum_{n\le x} \Lambda(n)\mu(n+h)\big)$, which is the right-hand side of Theorem 3 with E_3 absorbed into the $o(1)$. Their constant $A_h = \prod_{p\nmid h,\, p>2}(1-1/(p(p-1)))$ is (3). Huang–Li [2] transpose that chain to the Goldbach shift, and their (22) is its endpoint; the Goldbach-side precursor is Pan [11], and [6] develops the equidistribution side.

Three things are added here and none of them is the chain. The residual is *named* rather than absorbed, so the statement is an equality with an explicit error term instead of an asymptotic. It is *unconditional* in the Corollary-1 regime, which is possible only because Theorem 1 discharges the other consumption of EH_μ — [5] spend their hypothesis on S_4 by partial summation and [2] spend it on E_4 through their Lemma 5, and neither step is needed. And the three-way distinction of Note 4 is visible only once the $o(1)$ is replaced by a named object.

Closure at the level of identities is not closure at the level of *strength*, and the next two propositions separate the two. Every condition on E_3 unwinds, through Theorem 3, to a condition on $\tilde{r}(N)$ and $C(N)$ — that is the identity part, and it is permanent. But Huang–Li consume E_3 two-sidedly and at a saving of every power of log, and Goldbach needs neither.

Proposition 7 (the demand is one-sided, and its threshold is not a log power)**.** *In the Corollary-1 regime, binary Goldbach for all sufficiently large even N follows from*

$$E_3(\alpha) \;>\; -(1-\varepsilon)\,\mathfrak{S}(N)\big(1-\mathfrak{A}(N)\big)N \tag{7}$$

for a single fixed $\varepsilon > 0$, with $\mathfrak{A}(N)$ as in (3). The quantifier matters and is the one Proposition 8 uses: with an unquantified $1+o(1)$ in place of $1-\varepsilon$ the hypothesis is not a closed condition, and on the thin family of Note 21, where the threshold is $\asymp N/\log N$, it would no longer dominate the error terms below. This is strictly weaker than $E_3 \ll_A N(\log N)^{-A}$ in two independent ways: it constrains E_3 from one side only, and its threshold is $\mathfrak{S}(N)(1-\mathfrak{A}(N))N$, which is $\asymp N$ over the even N measured — median 0.333459 of N, and bounded below on all but the primorial shapes — and, on the family where it is thinnest, is $\asymp N/\log N$ (Note 21, Measurement 22).

Proposition 8 (the demand needs no saving in $\log N$). *Put*

$$B_\mu(N) \;:=\; \sum_{\substack{k<K \\ (k,N)=1}} \mu^2(k)\,(\log k)\,\bigl|E_\mu(N;k)\bigr|, \tag{8}$$

the restriction to squarefree k being what the triangle inequality on (4) leaves, since $\mu(k) = 0$ elsewhere. The letter is [2]'s, who write it without a subscript; the subscript is added here for the reason given for $\mathfrak{A}$ at (3) — one symbol must not carry two meanings — since the mean-term factor $B(K)$ of (11) is a different function of a smaller argument. This is sharper than the quantity [2] actually reach before appealing to EH_μ, namely $\log N \sum_{k<K,(k,N)=1} |E_\mu(N;k)|$, in which $\log k$ has been raised to $\log N$; the proposition below holds for either, and a fortiori for theirs. Then binary Goldbach for all sufficiently large even N follows from

$$B_\mu(N) \;\le\; (1-\varepsilon)\,\mathfrak{S}(N)\bigl(1-\mathfrak{A}(N)\bigr)N \tag{9}$$

for a single fixed $\varepsilon > 0$. Since $\mathfrak{S}(N)(1-\mathfrak{A}(N))$ is bounded away from 0 except on the thin family of Note 21, (9) is a constant-factor *bound — absolute values kept, and no saving in $\log N$ whatsoever — where $EH_\mu(N^{\theta'})$ asks for $\ll_A N(\log N)^{-A}$ for every A. The two are not bounds on the same object: $B_\mu(N)$ fixes $y = N$ and the class $a \equiv N$, restricts to squarefree k with $(k,N) = 1$, and carries the weight $\log k$, where (1) maximises over y and over the classes first. The comparison runs one way — each summand of $B_\mu(N)$ is at most $\log K$ times the corresponding summand of (1), so $EH_\mu(N^{\theta'})$ implies (9) and not conversely — and what (9) lowers is the strength, not the level.*

1.4 What is not claimed

- **No progress toward Goldbach.** Theorem 1 is unconditional but Goldbach-neutral: it removes exactly the half of the demand that carries no Goldbach content. The net progress toward the Goldbach conjecture is zero.

- **No new estimate for $C(N)$.** Theorem 3 places the whole surviving demand on $C(N) = \sum_{n<N} \Lambda(n)\mu(N-n)$; nothing here bounds that quantity.

- **No claim of novelty of mechanism.** The ingredients of Theorem 1 are classical (Section 8). Its difficulty is bookkeeping, and the two places where the bookkeeping bites are isolated in Section 2 and Note 12.

- **The weakenings do not reopen the design space.** Propositions 7 and 8 lower the required strength but not the required level; the divisor-switch route remains closed for every weight whose transform is accessible to Bombieri–Vinogradov [9]. The polynomial weights $w_k = f(\log k)$ lie outside that hypothesis, and what [9] establishes for them is weaker: their complete part is nonnegative and not known to be $o(N)$ — at degree one it is the binary Goldbach sum itself — and the tuning that would cancel it is circular. Neither closure is proved here. Both are [9]'s, and what this paper asserts is only the narrower thing it can: that Propositions 7 and 8 lower the required strength without lowering the required level, so that they leave the hypothesis under which [9] proves them in place. A reader who wants the design space closed must go to [9] for it.

- **The defect report is not a retraction.** [2]'s Theorem 1 and Corollary 1 stand as stated; what Section 7 supplies is a missing term and its treatment.

2 Notation and the mechanism

p and q always denote primes. φ is Euler's function, τ_3 the 3-fold divisor function, $\mathrm{rad}(u) = \prod_{p|u} p$. Constants $c, c_1, \ldots$ are positive and absolute unless subscripted; implied constants may depend on A and θ'. The letter A is reserved for the free exponent of (1); the local factor of (3) is $\mathfrak{A}$.

The mechanism of Theorem 1 is the following. The residue class in (6) is the fixed class $n \equiv N \pmod{k}$, i.e.

$$n \equiv N \pmod{k} \quad \iff \quad k \mid N - n.$$

So the k-sum is a divisor sum over $u = N - n$, namely the incomplete sum $\sigma_K(u) = \sum_{k|u,\ k<K,\ (k,N)=1} \mu(k)$, and the mechanism has three steps rather than one.

First one **completes** it. By Lemma 10, for squarefree u the complete sum $\sum_{k|u,\,(k,N)=1} \mu(k)$ equals $\mathbf{1}_{\mathrm{rad}(u)|N}$, which forces $u \mid \mathrm{rad}(N)$ and so leaves only $N^{o(1)}$ terms. *Then* the work sits entirely in the complementary sum, over $k \geq K$ — and it is only on that range that writing $u = mk$ gives

$$m \;=\; u/k \;<\; N/K \;=\; N^{1-\theta'} \;\leq\; N^{1/2-\delta}.$$

Finally, for squarefree $u = mk$ the factorisation forces $(m, k) = 1$ and $\mu(u)\mu(k) = \mu(m)\mu^2(k)$: the Möbius factor on the *long* variable k cancels against itself and leaves the nonnegative μ^2, while the surviving Möbius sits on the *short* variable m. That is the classical "Möbius on the short variable plus Bombieri–Vinogradov" configuration.

The completion step is load-bearing and easy to omit. Over the range $k < K$ as it stands the cofactor u/k runs up to u itself, so the bound $m < N^{1-\theta'}$ is simply false there and the surviving Möbius would sit on the *long* variable — the configuration for which no machine is known.

The words *short* and *long* describe the Corollary-1 regime and not what the proof uses. By Note 9 the only thing the argument asks of the cofactor is that its size lie below the level of distribution; under a level above 1/2 the same proof runs with m the longer of the two. Steps 1–3 of Section 3 carry out the argument in that order.

Note 9 (where $\theta' > 1/2$ is used, and which endpoint it is). Writing each line of Section 3 as an exponent pair $N^{a_0+a_1\theta'}(\log N)^{b_0+b_1A}$, and each appeal to Bombieri–Vinogradov as a constraint on the level rather than as a term, exactly one line of the twelve fails somewhere in $(0, 1)$: it is (19), the requirement that the modulus $q = m\,\mathrm{lcm}(d^2, e) \leq MD_0^2E_0 = N^{1-\theta'}e^{3\sqrt{\log N}}$ stay under what Bombieri–Vinogradov admits. The other eleven are the subtracted mean term of Step 0; $P(t)$; the four truncation-tail lines, being the progression part and the counting part of each of the d- and the e-tail; the two degeneracy discards of Lemma 11, the one at (16) and the one at (18); the truncation error the density of Lemma 15 carries into Proposition 17; the tail of Lemma 16; and the main term of Proposition 17. Each of the eleven is admissible on all of $(0, 1)$.

The $\log k$ branch of Section 5 adds no line that fails anywhere in $(0, 1)$, and it is worth saying why rather than repeating the accounting for it. On the residual the twelve carry over: the extra factor (22) moves b_0 and never a_0 or a_1, the moduli are the moduli of (18), so the appeal to Bombieri–Vinogradov there is the one already counted and not a second one, and the one term its Abel summation adds — the $w_m(1)$ per class — has the profile $N^{1-\theta'+o(1)}$ already on the list, as the degeneracy discard at (18) has. Two of its lines are new in kind, and they come from the completion rather than from the residual: the prime-power error of (21), which is $N^{1/2+\varepsilon}$, and the discard of the terms with $(u, N) > 1$, which is $N^{o(1)} \log^2 N$. Neither carries θ' at all, so each is admissible on the whole of $(0, 1)$ and neither moves the threshold. All five of the branch's lines — those two, the $w_m(1)$ term, the density tail, and the level call — are entered in the same table as the twelve and the six of Proposition 25, twenty-three rows in all, and the run behind this note adjudicates them.

The sixth of Proposition 25's rows is the b-tail discard of the bridge terms, a line of that proof distinct from its level call on the moduli. It was absent from the table until the estimate in Section 7 was corrected to count on the pair (m,k); counted that way it is $N(\log N)^{-(A+3)}(\log\log N)^2$, carries no θ', and is admissible on all of $(0,1)$. Counted on the modulus alone it is not, and it would then be an occurrence of $1/2$ inside a count of pairs, which no level of distribution moves — which is why the correction there and the row here are the same fact.

So the condition is $1-\theta' < \theta_E$, with θ_E the level of distribution of Λ in progressions. Replacing Bombieri–Vinogradov's $\theta_E = 1/2$ by any other value moves the threshold to exactly $1-\theta_E$ and moves no other line. The $1/2$ in Theorem 1 is thus a property of the input and not of the theorem: it is the level of distribution, carried into the proof by the size $N^{1-\theta'}$ of the cofactor the completion leaves, and by nothing else. Under EH the same proof gives Theorem 1 for every $\theta' > \varepsilon$.

The two endpoints of $(1/2,1)$ are not the same kind of thing. What fails at $\theta' = 1$ is a different and disjoint set — the tail of Lemma 16, Proposition 17, and the $m=1$ bridge term of Proposition 25 — and each fails there because $M = N^{1-\theta'}$ ceases to be a power of N, so that the configuration this section describes stops existing. The lower endpoint is an inequality between two exponents and moves when the input moves; the upper endpoint is where the parametrisation degenerates and does not move at all.

The threshold is also asymptotic in a way the statement hides. With the truncations of (10) the least θ' for which $MD_0^2E_0$ stays below $N^{1/2}$ is $1/2+3/\sqrt{\log N}$ — 0.737171 at $\log N = 160$ and 0.583853 at $\log N = 1280$, at least 1 — and hence admitting nothing — only until $\log N = 37$. The interval $(1/2,1)$ is the limit of a family of finite-N intervals, none of which contains its own endpoint, and no N at which anything in this paper is measured lies in any of them.

That boundary is where the choice of truncation shows. A truncation by a power of $\log N$ puts it at $\log N = 215$ for $A = 3$ alone and further out for every larger A, the level then carrying $(\log N)^{4A+8}$; the exponential truncation carries $e^{3\sqrt{\log N}}$ instead, which is $N^{o(1)}$ and independent of A.

The run behind this note has two of its registered predictions refuted, and the account of the two endpoints given above is neither of them. The first asked which lines of the proof supply the *upper* endpoint of $(1/2,1)$, and the answer is that none does: no line fails anywhere strictly inside $(0,1)$ except the level. The premise was wrong rather than the answer, and what is printed above is the replacement question, registered before it was run.

The second refutation is the replacement question's own, and it arrived when the $\log k$ branch was entered in the table. That prediction named the three lines it would tolerate failing inside $(0,1)$ and declared itself refuted at a fourth; the branch's level call is a fourth occurrence of the level, so the count reached four. Its sentence — no line fails inside $(0,1)$ except the level constraint — is what the table shows, at four occurrences as at three. The rule counts occurrences where the sentence counts constraints, and it was fixed before the run, so it is reported and not rewritten (`audit_theta_halfline.py`).

3 Proof of Theorem 1

Fix $A > 0$ and set

$$D_0 \;=\; E_0 \;:=\; \exp\bigl(\sqrt{\log N}\bigr). \tag{10}$$

One truncation serves every A. A truncation by a power of $\log N$ would have to grow with A and — more to the point — would leave the tails below at exactly $N(\log N)^{-A}$, the size Bombieri–Vinogradov itself contributes, so that the second sentence of Theorem 1 would be false. The exponential choice puts every ingredient but Bombieri–Vinogradov below every power of $\log N$, at no cost to (19).

3.1 Step 0: the subtracted mean term

By (6),

$$T_1(t) \;=\; D(t) \;-\; C(t)\,B(K), \qquad D(t) := \sum_{\substack{k<K\\(k,N)=1}} \mu(k) \sum_{\substack{n\le t\\ n\equiv N\ (k)}} \Lambda(n)\mu(N-n), \tag{11}$$

where $B(K) = \sum_{k<K,(k,N)=1} \mu(k)/\varphi(k)$. Huang–Li's Lemma 1 (a form of Goldston–Yıldırım [1]) gives $B(K) \ll e^{-c_1\sqrt{\log K}}$, and trivially $|C(t)| \le \sum_{n<N} \Lambda(n) \ll N$. Hence

$$|C(t)B(K)| \;\ll\; Ne^{-c\sqrt{\log N}} \tag{12}$$

uniformly in t.

3.2 Step 1: divisor switching (an exact identity)

Since $n \equiv N \pmod k$ with $n \le t < N$ is the same as $u := N-n$ being a multiple of k in $[N-t, N)$,

$$D(t) \;=\; \sum_{N-t\le u<N} \Lambda(N-u)\,\mu(u)\,\sigma_K(u), \qquad \sigma_K(u) := \sum_{\substack{k\mid u,\ k<K\\(k,N)=1}} \mu(k). \tag{13}$$

This is a finite rearrangement, hence exact. It is verified independently by a script that accumulates the two sides in genuinely different index orders — the k-side by modular slices, the u-side against a divisor-sum array — and finds them equal to 10^{-16} relative to N (`audit_switch_identity.py`).

3.3 Step 2: the complete divisor sum

Lemma 10. *For every $u \ge 1$,* $\displaystyle\sum_{k\mid u,\ (k,N)=1} \mu(k) = \mathbf{1}_{\mathrm{rad}(u)\mid N}$.

Proof. The sum is multiplicative in u with local factor 1 at $p \mid N$ and $1-1=0$ at $p \nmid N$. Hence it vanishes unless every prime of u divides N. □

Splitting $\sigma_K(u)$ into the complete sum minus the tail $k \ge K$, (13) becomes $D(t) = P(t) - R(t)$ with

$$P(t) = \sum_{\substack{N-t\le u<N\\ \mathrm{rad}(u)\mid N}} \Lambda(N-u)\mu(u), \qquad R(t) = \sum_{N-t\le u<N} \Lambda(N-u)\mu(u) \sum_{\substack{k\mid u,\ k\ge K\\(k,N)=1}} \mu(k). \tag{14}$$

In $P(t)$ the conditions $\mu(u) \ne 0$ and $\mathrm{rad}(u) \mid N$ force $u \mid \mathrm{rad}(N)$, so there are at most $2^{\omega(N)} = N^{o(1)}$ terms, each $\ll \log N$:

$$P(t) \ll N^{o(1)}. \tag{15}$$

3.4 Step 3: the residual, and the degeneracy lemma

Write $u = mk$ with $k \geq K$, so $m = u/k < N/K =: M$. For squarefree u the factorisation forces $(m,k) = 1$ and $\mu(u)\mu(k) = \mu(m)\mu^2(k)$, so

$$R(t) = \sum_{m<M} \mu(m) \sum_{\substack{k \geq K,\ (k,m)=1,\ (k,N)=1 \\ N-t \leq mk < N}} \mu^2(k)\, \Lambda(N - mk). \tag{16}$$

Lemma 11 (degeneracy)**.** *The contribution to (16) of the terms with $(k,N) > 1$ is $O(N^{o(1)})$. The same bound holds for the terms in which the modulus constructed in Step 4 is not coprime to N.*

Proof. Suppose $p \mid (k,N)$ and $\Lambda(N - mk) \neq 0$, say $N - mk = q^\ell$. Since $p \mid k$ and $p \mid N$ we get $p \mid N - mk$, hence $q = p$ and $n := N - mk = p^\ell$ with $p \mid N$. The number of such $n < N$ is $\leq \sum_{p \mid N} \log N / \log p \ll (\log N)^2$; for each, the pairs (m,k) with $mk = N - n$ number $\leq \tau(N - n) \ll N^{o(1)}$, and each term is $\ll \log N$.

For the second sentence, let $q = m \operatorname{lcm}(d^2, e)$ be the modulus built in Step 4 and suppose $g = (q,N) > 1$. A term of (18) in that class has $n \equiv N \pmod q$ with $1 \leq n \leq \mathcal{T}_m(t)$, so $g \mid N - n$; together with $g \mid N$ this gives $g \mid n$, and any $p \mid g$ then divides n, so $\Lambda(n) \neq 0$ forces $n = p^\ell$ with $p \mid N$. The number of such $n < N$ is again $\leq \sum_{p \mid N} \log N / \log p \ll (\log N)^2$. For each such n, a triple (m,d,e) places it in a degenerate class only if $m \operatorname{lcm}(d^2,e) \mid N - n$, so each of m, d^2 and e divides $N - n$ and the triples number $\ll \tau(N-n)^3 \ll N^{o(1)}$; with $\Lambda(n) \ll \log N$ the total is $\ll N^{o(1)}$. $\square$

By Lemma 11 we may drop the condition $(k,N) = 1$ from (16) at a cost of $O(N^{o(1)})$. This is essential: expanding $\mathbf{1}_{(k,N)=1}$ by Möbius over $e \mid N$ would multiply the number of Bombieri–Vinogradov calls by $2^{\omega(N)} = N^{o(1)}$ and destroy the budget.

3.5 Step 4: unfolding μ^2 and the coprimality

Insert $\mu^2(k) = \sum_{d^2 \mid k} \mu(d)$ and $\mathbf{1}_{(k,m)=1} = \sum_{e \mid (k,m)} \mu(e)$ and truncate at D_0, E_0 of (10).

Tail $d > D_0$. Such terms have $d^2 \mid k$, hence $n = N - mk \equiv N \pmod{md^2}$; the number of $n \leq N$ in that progression is $\ll N/(md^2) + 1$ and each $\Lambda \ll \log N$. Summing,

$$\ll \log N \sum_{m<M} \sum_{d > D_0} \Big(\frac{N}{md^2} + 1 \Big) \ll \frac{N (\log N)^2}{D_0} + \sqrt{NM} \log N,$$

which is

$$\ll N (\log N)^2 e^{-\sqrt{\log N}} + N^{1-\theta'/2} \log N.$$

Tail $e > E_0$. Such terms have $e \mid k$ and $e \mid m$, hence $n \equiv N \pmod{me}$, and

$$\ll \log N \sum_{m<M} \sum_{\substack{e \mid m \\ e > E_0}} \Big(\frac{N}{me} + 1 \Big) \ll \frac{N \log N}{E_0} \sum_{m<M} \frac{\tau(m)}{m} + M \log M \log N,$$

which is

$$\ll N (\log N)^3 e^{-\sqrt{\log N}} + M (\log N)^2.$$

Both are $\ll N e^{-c\sqrt{\log N}}$ for any $c < 1$, hence below every power of $\log N$. In the remaining range put $L := \operatorname{lcm}(d^2, e)$ and $q := mL$; the conditions $d^2 \mid k$, $e \mid k$ and $n = N - mk$ give $n \equiv N \pmod q$, and $mk < N$, $mk \geq \max(N - t, mK)$ give $1 \leq n \leq \mathcal{T}_m(t)$ where

$$\mathcal{T}_m(t) := \min\big(t,\ N - mK\big). \tag{17}$$

Therefore, with $\psi(y;q,a)=\sum_{n\le y,\,n\equiv a\,(q)}\Lambda(n)$,

$$R(t)=\sum_{m<M}\mu(m)\sum_{d\le D_0}\mu(d)\sum_{\substack{e|m\\ e\le E_0}}\mu(e)\,\psi\big(\mathcal{T}_m(t);\,q,\,N\big)\;+\;O\Big(Ne^{-c\sqrt{\log N}}\Big).\tag{18}$$

The error here is the two truncation tails just bounded, and it is exponential: Bombieri–Vinogradov has not been applied yet, and the second sentence of Theorem 1 — that every ingredient but Bombieri–Vinogradov saves exponentially — is read off this line among others. By Lemma 11 we may further restrict to $(q,N)=1$ at a cost of $O(N^{o(1)})$; the lemma reaches that bound by counting the n, of which there are $\ll(\log N)^2$, rather than the triples. The cruder count — each degenerate class contributes $N^{o(1)}$ and there are $\ll MD_0E_0\tau$-many triples — gives only $\ll N^{1-\theta'+o(1)}$. That is weaker, but it already suffices at every $\theta'\in(0,1)$, and it is the form in which the exponent bookkeeping of Note 9 records this line.

Note 12 (the false-positive trap). Restricting the *main terms* to classes with $(q,N)=1$ is not cosmetic. For the m of Lemma 15 — squarefree and coprime to N, with $e\mid m$ — the condition $(q,N)=1$ is exactly the condition $(d,N)=1$, and it is what makes the local factor of that lemma equal 1 at $p\mid N$ instead of $1-1/(p(p-1))$. Assigning a main term $\mathcal{T}_m/\varphi(q)$ to a degenerate class puts those factors back, and so multiplies the density of the whole computation by $\prod_{p|N}\big(1-\frac{1}{p(p-1)}\big)$: 0.5000000 at $N=2^{20}$, 0.3790428 at $N=510\,510$, 0.3779345 at $N=9\,699\,690$. Carried into the $\log k$ branch of Section 5, where $\mathfrak{A}(N)\widetilde{G}(1)=-\mathfrak{S}(N)$ is the identity everything there rests on, that factor puts the constant out by between 0.4999998 and 0.6220658 of $\mathfrak{S}(N)$ across the even N measured, and the branch reads the difference as an apparent *refutation* of EH_μ. It is recorded here because it is the most plausible false positive in this circle of ideas.

The factor is easy to get wrong in a particular way, and the run below registers the wrong value as its own prediction W1 so that the error is on record rather than merely avoided: $N/\varphi(N)$ is above 1 at every even N while the shift is below 1 at every even N, so the two agree neither in size nor in direction, and a reader who reaches for the density of the coprime classes reaches for the wrong one (`audit_trap_density.py`).

Note the size of the modulus: $q=m\,\mathrm{lcm}(d^2,e)\le MD_0^2E_0$, so by (10)

$$q\;\le\;N^{1-\theta'}e^{3\sqrt{\log N}}\;=\;N^{1/2-\delta}e^{3\sqrt{\log N}}\;\le\;N^{1/2-\delta/2}\tag{19}$$

for N large. This is the level at which Bombieri–Vinogradov is applied.

3.6 Step 5: Bombieri–Vinogradov

Lemma 13 (τ-weighted Bombieri–Vinogradov)**.** *For all $A,B>0$ there is $C=C(A,B)$ such that for $Q\le N^{1/2}(\log N)^{-C}$,*

$$\sum_{q\le Q}\tau_3(q)^B\max_{y\le N}\max_{(a,q)=1}\Big|\psi(y;q,a)-\frac{y}{\varphi(q)}\Big|\;\ll_{A,B}\;\frac{N}{(\log N)^A}.$$

Proof. Write $\mathcal{E}_q$ for the inner maximum. By Cauchy–Schwarz,

$$\sum_q\tau_3^B\mathcal{E}_q\;\le\;\Big(\sum_q\tau_3^{2B}\mathcal{E}_q\Big)^{1/2}\Big(\sum_q\mathcal{E}_q\Big)^{1/2}.$$

For the first factor use the trivial bound $\mathcal{E}_q\ll(N/\varphi(q))\log N$, giving $\ll N(\log N)^{C_1(B)}$; for the second use Bombieri–Vinogradov [8] with exponent $A'=2A+C_1(B)$. That theorem is stated for $N^{1/2}(\log N)^{-A'}\le Q'\le N^{1/2}$, a window closed below as well as above, whereas the Q here lies beneath it. The gap is closed by monotonicity: the summand is non-negative, so for $Q \le Q'$ one has $\sum_{q\le Q} \mathcal{E}_q \le \sum_{q\le Q'} \mathcal{E}_q$, and taking $Q' = N^{1/2}(\log N)^{-(A'+3)}$ and applying the theorem with the exponent $A'+3$ in place of A' — for which that Q' is the lower endpoint of the window and not a point beneath it — gives $\ll N(\log N)^{-(A'+3)} \le N(\log N)^{-A'}$. The constant promised in the statement is $C = A' + 3 = 2A + C_1(B) + 3$. $\square$

Note 14 (the constant is not effective). The implicit constant in Bombieri–Vinogradov is non-effective, the proof appealing to the Siegel–Walfisz theorem [8]. The constants implied by $\ll_{A,\theta'}$ in Theorem 1, and hence in Corollary 2, inherit that: they are not effectively computable, and no statement below is asserted with an effective constant.

A modulus q arises in (18) from at most $\sum_{m|q} \tau(q/m)^2 \le \tau(q)^3 \le \tau_3(q)^3$ triples (m, d, e): given $m \mid q$ there are at most $\tau(q/m)$ choices of d with $d^2 \mid q/m$ and at most $\tau(q/m)$ of e with $\mathrm{lcm}(d^2, e) = q/m$. Writing $\psi(\mathcal{T}_m; q, N) = \mathcal{T}_m/\varphi(q) + \mathcal{E}$ and applying Lemma 13 with $B = 3$ and (19),

$$R(t) \;=\; \mathrm{MT}(t) \;+\; O\!\left(\frac{N}{(\log N)^A}\right), \qquad \mathrm{MT}(t) = \sum_{\substack{m<M\\(m,N)=1}} \mu(m)\,\mathcal{T}_m(t)\,c_{D_0,E_0}(m), \tag{20}$$

where $c_{D_0,E_0}(m) = \sum_{d\le D_0} \sum_{e|m,\, e\le E_0} \mu(d)\mu(e)\mathbf{1}_{(d,N)=1}/\varphi(m\,\mathrm{lcm}(d^2, e))$.

3.7 Step 6: the density

Lemma 15. *Let m be squarefree with $(m, N) = 1$ and put*

$$c(m) := \sum_{d\ge1} \sum_{e|m} \frac{\mu(d)\mu(e)\mathbf{1}_{(d,N)=1}}{\varphi(m\,\mathrm{lcm}(d^2, e))}.$$

Then $c(m) = \mathfrak{A}(N)\lambda(m)/m$ with $\mathfrak{A}(N)$ as in (3) and $\lambda(m) = \prod_{p|m}\bigl(1 - \frac{1}{p(p-1)}\bigr)^{-1}$. Moreover $c_{D_0,E_0}(m) = c(m) + O(1/(\varphi(m)D_0))$.

Proof. The sum is multiplicative. If $p \nmid m$ then $p \nmid e$ and the local factor is $1 - 1/\varphi(p^2) = 1 - 1/(p(p-1))$ for $p \nmid N$, and 1 for $p \mid N$. If $p \mid m$ (so $p \nmid N$) the p-part of $m\,\mathrm{lcm}(d^2, e)$ is $p^{1+\max(2v_p(d),v_p(e))}$, and the four choices $(v_p(d), v_p(e)) \in \{0, 1\}^2$ contribute

$$\frac{1}{p-1} - \frac{1}{p(p-1)} - \frac{1}{p^2(p-1)} + \frac{1}{p^2(p-1)} \;=\; \frac{1}{p}.$$

Hence $c(m) = \prod_{p|m} p^{-1} \cdot \prod_{p\nmid m,\ p\nmid N}(1 - \frac{1}{p(p-1)}) = m^{-1}\mathfrak{A}(N)\lambda(m)$. The truncation error is the tail $\sum_{d>D_0} 1/\varphi(md^2) \ll 1/(\varphi(m)D_0)$ and similarly for e. $\square$

The load-bearing line is the local factor. Equivalently, for squarefree m,

$$\sum_{g|m} \frac{\mu(g)}{\varphi(m/g)\,g\,\varphi(g)} = \frac{1}{m}:$$

the local factor is exactly p^{-1}, so the density exponent is exactly 1 and $1/\zeta$ occurs to the *first* power in Lemma 16 below. A non-integral exponent would give, by the Selberg–Delange method, only $(\log x)^{-c}$ for a fixed c in Lemma 16, and Proposition 17 would then give $\mathrm{MT} \ll N(\log N)^{-c}$ with c fixed, which the proof below cannot use. That is a statement about *this* argument and not about Theorem 1, and the method is named only to say what the alternative would look like: nothing below appeals to it. The identity is verified in exact rational arithmetic for all squarefree $m < 400$, with zero mismatches (`audit_density_identity.py`).

Lemma 16. *Let $f(m) := \mu(m)\lambda(m)\mathbf{1}_{(m,N)=1}$ and $G(x) := \sum_{m\le x} f(m)/m$. Then*

$$G(x) \;\ll\; \frac{N}{\varphi(N)}\, e^{-c\sqrt{\log x}} + x^{-1/4}e^{C\sqrt{\log N}} \qquad (x \ge 2).$$

Proof. Write $F(s) = \sum_m f(m)m^{-s} = \prod_{p\nmid N}(1-\lambda(p)p^{-s})$ and $F(s) = \zeta(s)^{-1}H(s)$, so $f = \mu * h$ with h multiplicative, $h(p^j) = 1$ for $p \mid N$ and $h(p^j) = 1-\lambda(p) = O(p^{-2})$ for $p \nmid N$. Then $G(x) = \sum_{b\le x}\frac{h(b)}{b}\sum_{a\le x/b}\frac{\mu(a)}{a}$. For $b \le \sqrt{x}$ use $\sum_{a\le y}\mu(a)/a \ll e^{-c\sqrt{\log y}}$, which is the classical zero-free region estimate [7], together with $\sum_b |h(b)|/b \ll \prod_{p|N}(1-1/p)^{-1} \ll N/\varphi(N)$. For $b > \sqrt{x}$ bound $\sum_{b>y}|h(b)|/b \le y^{-1/2}\sum_b |h(b)|b^{-1/2} \ll y^{-1/2}\prod_{p|N}(1-p^{-1/2})^{-1} \ll y^{-1/2}e^{C\sqrt{\log N}}$. □

Since $M = N^{1-\theta'}$ is a fixed power of N, the second term of Lemma 16 is negligible at $x \asymp M$, and $N/\varphi(N) \ll \log\log N$; so $G(x) \ll e^{-c'\sqrt{\log x}}$ for x of that size. It is not so for small x: at $x = 2$ the second term is $2^{-1/4}e^{C\sqrt{\log N}}$, and the bound $G(x) \ll e^{-c'\sqrt{\log x}}$ fails on the whole of $x \le e^{4C\sqrt{\log N}}$. Proposition 17 integrates G over $[1, M]$ and therefore splits there; the split is carried out in its proof.

3.8 Step 7: the main term dies, uniformly in t

Proposition 17. $\sup_{1\le t<N} |\mathrm{MT}(t)| \ll Ne^{-c\sqrt{\log N}}$.

Proof. By Lemma 15, $\mathrm{MT}(t) = \mathfrak{A}(N)\sum_{m<M}\frac{f(m)}{m}\mathcal{T}_m(t) + O(N\log N/D_0)$ with $\mathcal{T}_m(t)$ as in (17). Put $m_0 := (N-t)/K$, so $\mathcal{T}_m(t) = t$ for $m \le m_0$ and $\mathcal{T}_m(t) = N - mK$ for $m > m_0$; in particular $x \mapsto \mathcal{T}_x(t)$ is non-increasing, is constant on $[1, m_0]$, has derivative $-K$ on (m_0, M), and $\mathcal{T}_M(t) = 0$. Abel summation against G of Lemma 16 gives

$$\sum_{m<M}\frac{f(m)}{m}\mathcal{T}_m(t) = G(M)\,\mathcal{T}_M(t) + K\int_{m_0}^{M} G(x)\,dx = K\int_{m_0}^{M} G(x)\,dx.$$

Hence, uniformly in t,

$$\Big|\sum_{m<M}\frac{f(m)}{m}\mathcal{T}_m(t)\Big| \le K\int_1^M |G(x)|\,dx.$$

Lemma 16's second term is at least 1 below $x_0 := e^{4C\sqrt{\log N}}$, so the integral is split there. On $[1, x_0]$ use the trivial $|G(x)| \ll \log x$, giving $\ll Kx_0\log x_0 \ll N^{\theta'}e^{O(\sqrt{\log N})}$. On $[x_0, M]$ the two terms of Lemma 16 give

$$K\int_{x_0}^{M}\Big(\frac{N}{\varphi(N)}e^{-c\sqrt{\log x}} + x^{-1/4}e^{C\sqrt{\log N}}\Big)dx \ll KMe^{-c'\sqrt{\log M}} + KM^{3/4}e^{C\sqrt{\log N}},$$

and with $KM = N$, $\log M \asymp \log N$ and $M = N^{1-\theta'}$ the two are $\ll Ne^{-c''\sqrt{\log N}}$ and $\ll N^{(3+\theta')/4}e^{C\sqrt{\log N}}$ respectively. All three pieces are $\ll Ne^{-c\sqrt{\log N}}$ for $\theta' < 1$, the first and the third being powers of N below N itself. The truncation error carried by Lemma 15 is $\ll \sum_{m<M}|\mathcal{T}_m(t)|/(\varphi(m)D_0)$, and $\mathcal{T}_m(t) \le N$, so with $\sum_{m<M}1/\varphi(m) \ll \log M$ it is $\ll N\log N/D_0 = N\log N\, e^{-\sqrt{\log N}} \ll Ne^{-c\sqrt{\log N}}$ for every $c < 1$. The size here is N and not M: the truncation is per term of the harmonic sum, and the terms are of the size of N. □

The cancellation is genuine and global: the term $m = 1$ alone contributes $\mathcal{T}_1(t) \asymp N$ — the endpoint of (17), not the weighted sum of Theorem 1 — so the smallness of MT is entirely due to the cancellation in $\sum f(m)/m$.

3.9 Conclusion

Combining (11), (12), (15), (20) and Proposition 17:

$$T_1(t) = \underbrace{P(t)}_{N^{o(1)}} - \underbrace{\mathrm{MT}(t)}_{\ll Ne^{-c\sqrt{\log N}}} - \underbrace{O\big(N(\log N)^{-A}\big)}_{\mathrm{BV}} - \underbrace{C(t)B(K)}_{\ll Ne^{-c\sqrt{\log N}}},$$

uniformly in $t < N$. This proves Theorem 1, and exhibits Bombieri–Vinogradov as the only ingredient that does not give an exponential saving. The bound cannot be improved to $Ne^{-c\sqrt{\log N}}$ by this argument: Bombieri–Vinogradov enters only as the black box of Lemma 13, which yields $N(\log N)^{-A}$ for each fixed A and nothing beyond it, so a saving the box does not give cannot be recovered here. That the box gives no more is a property of its standard proofs, whose Siegel–Walfisz input is restricted to $q \leq (\log N)^B$ with B fixed; that is a statement about [8] and is not proved here. □

4 Proof of Corollary 2

Fix k and set $a_n := \Lambda(n)\mu(N-n)\big(\mathbf{1}_{n\equiv N\,(k)} - 1/\varphi(k)\big)$, so that $E_\mu(t;k) = \sum_{n\leq t} a_n$ and the bracket in (5) is $\sum_{n<N} a_n \log(N-n)$. Since $\log(N-n)$ vanishes at $n = N-1$ and has derivative $-1/(N-t)$, Abel summation gives

$$\sum_{n<N} a_n \log(N-n) \;=\; \int_1^{N-1} \frac{E_\mu(t;k)}{N-t}\,dt.$$

The weight $\log(N-n)$ does not depend on k, so multiplying by $\mu(k)$ and summing over $k < K$ with $(k,N) = 1$ (a finite sum) gives the exact identity

$$E_4(\alpha) \;=\; \int_1^{N-1} \frac{T_1(t)}{N-t}\,dt.$$

Hence $|E_4(\alpha)| \leq \big(\sup_{t<N}|T_1(t)|\big)\log N \ll_A N(\log N)^{1-A}$ by Theorem 1. As A is arbitrary this is Corollary 2. Huang–Li's Lemma 5 is invoked only to bound E_4; with the above it is not needed, and the only remaining appearance of EH_μ in their §3 is through $E_3(\alpha)$. □

5 Proof of Theorem 3: the permanent closure

Take the weight $w_k = \log k$, i.e. the functional E_3 of (4). Steps 0–1 are unchanged, and the complete divisor sum of Lemma 10 is replaced by the following.

Lemma 18. *For every $u \geq 1$, write $u = u_N u'$ where u_N is the largest divisor of u composed of primes dividing N. Then $\sum_{k|u,\ (k,N)=1} \mu(k)\log k = -\Lambda(u')$.*

Proof. The sum is over $k \mid u'$ and equals $-\Lambda(u')$ by $\mu * \log = \Lambda$. □

Hence the complete piece of E_3 is

$$-\sum_{1\leq u<N} \Lambda(N-u)\,\mu(u)\,\Lambda(u') \;=\; -\sum_{u<N,\ (u,N)=1} \Lambda(N-u)\mu(u)\Lambda(u) + O(N^{o(1)}\log^2 N),$$

the discarded terms being those with $(u,N) > 1$, which are degenerate in the sense of Lemma 11: a prime $p \mid (u,N)$ divides $N-u$ as well, so $\Lambda(N-u) \neq 0$ forces $N-u$ to be a power of p, and the number of such u is $\ll (\log N)^2$ while each term is $\ll (\log N)^2$. The set discarded here looks like a set of size N and is not one; that it is not is the same mechanism as Lemma 11, applied to u instead of to k. Now $\mu(u)\Lambda(u)$ is supported on primes $u = p$, where it equals $-\log p$; prime powers $u = p^\ell$, $\ell \ge 2$, have $\mu(u) = 0$. So the complete piece equals

$$\sum_{p<N} \Lambda(N-p)\log p \;+\; O\big(N^{o(1)}\log^2 N\big) \;=\; \sum_{n<N} \Lambda(n)\Lambda(N-n) + O\big(N^{1/2+\varepsilon}\big), \tag{21}$$

the binary Goldbach sum itself. The door is free: divisor switching costs nothing, and immediately behind it stands the object one was trying to avoid.

Two further terms have to be evaluated, and neither vanishes.

(i) The residual main term. Steps 3–6 are repeated with the extra factor $\log k$, and what has to be checked is that the factor passes through each of them. It does, for one reason: on the residual range $k = (N-n)/m$, so that

$$\log k \;=\; w_m(n), \qquad w_m(n) := \log \frac{N-n}{m}, \tag{22}$$

a function of n and m alone. It does not involve d, e, or the modulus $q = m\,\mathrm{lcm}(d^2, e)$ built from them.

Steps 3 and 4 are therefore unchanged. The factorisation $u = mk$, the degeneracy lemma, and the unfolding of $\mu^2(k)$ and $\mathbf{1}_{(k,m)=1}$ all act on k through its divisors and not through its size, and w_m rides along as a coefficient. Two bookkeeping items move, each by one power of $\log N$: in Lemma 11 a term is now $\ll (\log N)^2$ rather than $\ll \log N$, which leaves that lemma's $O(N^{o(1)})$ where it was, and in the two truncation tails of Step 4 the factor $\log N$ standing for $\max \Lambda$ becomes $(\log N)^2$, so those tails read $\ll N(\log N)^3 e^{-\sqrt{\log N}} + N^{1-\theta'/2}(\log N)^2$ and $\ll N(\log N)^4 e^{-\sqrt{\log N}} + M(\log N)^2$, still $\ll N e^{-c\sqrt{\log N}}$ for every $c < 1$. The moduli are the same moduli, so (19) is untouched.

Step 5 is where the factor costs something, and it costs exactly one $\log N$. Fix m and write $\psi_w(y; q, a) = \sum_{n \le y,\ n \equiv a\,(q)} \Lambda(n) w_m(n)$. On $1 \le n \le \mathcal{T}_m(t)$ one has $N - n \ge mK$, so $w_m \ge \log K > 0$, and w_m decreases with $-w_m'(y) = 1/(N-y) \ge 0$; Abel summation gives

$$\psi_w(\mathcal{T}_m; q, N) \;=\; w_m(\mathcal{T}_m)\,\psi(\mathcal{T}_m; q, N) \;+\; \int_1^{\mathcal{T}_m} \frac{\psi(y; q, N)}{N-y}\,dy.$$

Replacing $\psi(y; q, N)$ by $y/\varphi(q)$ in both places costs at most

$$\Big(w_m(\mathcal{T}_m) + \int_1^{\mathcal{T}_m} \frac{dy}{N-y}\Big) \max_{y \le N} \Big|\psi(y; q, N) - \frac{y}{\varphi(q)}\Big| \;=\; w_m(1)\,\mathcal{E}_q \;\le\; (\log N)\,\mathcal{E}_q,$$

the bracket telescoping to $w_m(1) = \log((N-1)/m)$. Summing over the triples (m, d, e) exactly as in Step 5 and applying Lemma 13 with $B = 3$ and with $A + 1$ in place of A turns that into $O(N(\log N)^{-A})$. The one extra $\log N$ is absorbed by the arbitrariness of A, which is why no exponent moves.

What is left in place of $\mathcal{T}_m/\varphi(q)$ is $\varphi(q)^{-1}\big(w_m(\mathcal{T}_m)\mathcal{T}_m + \int_1^{\mathcal{T}_m} y(N-y)^{-1} dy\big) = \varphi(q)^{-1}\big(\int_1^{\mathcal{T}_m} w_m(y)\,dy + w_m(1)\big)$, again by parts; the $w_m(1)$ is $\ll \log N$ per class and, over the $\ll M D_0 E_0 N^{o(1)}$ triples, is $\ll N^{1-\theta'+o(1)}$. Step 6 is unchanged, the density $c(m)$ of Lemma 15 being a property of the modulus and not of the weight, and its truncation error contributes $\ll N \log N \sum_{m<M} (\varphi(m) D_0)^{-1} \ll N(\log N)^2 D_0^{-1}$. Substituting $v = N - y$ in $\int_1^{\mathcal{T}_m} w_m(y)\,dy$ and taking $t = N$, where $N - \mathcal{T}_m = mK$, the outcome is the main term $\mathfrak{A}(N) \sum_{m<M} \mu(m)\lambda(m) m^{-1} \int_{mK}^N \log(v/m)\,dv$. Substituting $v = mt$ evaluates the integral as $N\log(N/m) - N - mK\log K + mK$, of which only the $-N\log m$ piece survives the cancellation: Lemma 16 gives $G(x) = \sum_{m \le x} f(m)/m \ll e^{-c\sqrt{\log x}}$, which kills the $N\log N$ and the $-N$ piece, and partial summation against G gives $\sum_{m \le x} f(m) \ll x e^{-c\sqrt{\log x}}$, which kills the $-mK\log K$

and the mK piece; the same estimate bounds the tail $\widetilde{G}(M) - \widetilde{G}(1) \ll e^{-c\sqrt{\log M}}$, absorbed in the error term — where $\widetilde{G}(M)$ is the partial sum to M and $\widetilde{G}(1)$ is the value at $s = 1$ defined just below, not the partial sum to 1, which is 0. What is left is $-\mathfrak{A}(N)\, N\, \widetilde{G}(1)$ with $\widetilde{G}(1) = \lim_x \sum_{m\le x} \mu(m)\lambda(m)\mathbf{1}_{(m,N)=1} \log m/m$. The constant is fixed by

$$\mathfrak{A}(N)\, \widetilde{G}(1) \;=\; -\mathfrak{S}(N), \tag{23}$$

so that the residual main term is $+\mathfrak{S}(N)\, N + O(N(\log N)^{-A})$.

The sign in (23) is forced, and it is the whole identity: with $F(s) = \sum_m f(m)m^{-s} = \zeta(s)^{-1}H(s)$ as in Lemma 16 one has $\widetilde{G}(1) = -F'(1) = -H(1) < 0$, while $\mathfrak{A}(N)H(1) = \mathfrak{S}(N)$ identically — the local factors pair as

$$\Big(1 - \tfrac{1}{p(p-1)}\Big)\Big(1 - \tfrac{1}{(p^2-p-1)(p-1)}\Big) = 1 - \tfrac{1}{(p-1)^2}$$

at $p \nmid N$, and as $p/(p-1)$ at $p \mid N$. Numerically, $\mathfrak{A}(N)\widetilde{G}(x) = -1.760250$ at $x = 4 \cdot 10^6$ against $\mathfrak{S}(N) = 1.760432$ (`audit_E3_constant.py`).

(ii) The subtracted mean term. It carries the factor

$$B_{\log}(K) := \sum_{\substack{k<K\\(k,N)=1}} \frac{\mu(k)\log k}{\varphi(k)} \;=\; -\mathfrak{S}(N) + O\big(e^{-c\sqrt{\log K}}\log K\big),$$

which is exactly Huang–Li's Lemma 1: subtracting $\sum_{k\le K,\,(k,N)=1} \mu(k)\varphi(k)^{-1}\log(K/k) = \mathfrak{S}(N) + O(e^{-c_1\sqrt{\log K}})$ from $\log K \cdot \sum_{k\le K,\,(k,N)=1} \mu(k)/\varphi(k) = O(\log K\, e^{-c_1\sqrt{\log K}})$ gives the claim. The coprimality condition is carried throughout: without it the limiting constant is not $\mathfrak{S}(N)$. Independently: with $f(s) = \prod_{p\nmid N}(1 - \frac{p^{-s}}{p-1})$ one has $f(s) = \zeta(s+1)^{-1}h(s)$ with $h(0) = \mathfrak{S}(N)$, so $B_{\log}(\infty) = -f'(0) = -\mathfrak{S}(N)$. Since the mean term is $-C(N)B_{\log}(K)$ with $C(N) = \sum_{n<N} \Lambda(n)\mu(N-n)$, it contributes $+\mathfrak{S}(N)\, C(N)$.

Collecting (21), (i) and (ii):

$$E_3(\alpha) = \sum_{n<N} \Lambda(n)\Lambda(N-n) - \mathfrak{S}(N)N + \mathfrak{S}(N)C(N) + O_A\bigg(\frac{N}{(\log N)^A}\bigg),$$

which is the assertion of Theorem 3. Rearranging, the bound $E_3(\alpha) \ll_A N(\log N)^{-A}$ is therefore equivalent to

$$\tilde{r}(N) = \mathfrak{S}(N)\big(N - C(N)\big) + O_A\big(N(\log N)^{-A}\big)\,,$$

which is Huang–Li's equation (22). It is their (22) that is the equality here; the inequality (2) above is what they deduce from it, and the equivalence just proved is with the equality and not with the deduction. Given that, $\tilde{r}(N)$ is positive for large even N — binary Goldbach — because $|C(N)| \le \mathfrak{A}(N)N(1 + o(1))$ with $\mathfrak{A}(N) < 1$; and, since $\mathfrak{S}(N) \ge 2\prod_{p>2}(1 - (p-1)^{-2}) \gg 1$, $\tilde{r}(N) \sim \mathfrak{S}(N)N$ holds if and only if $C(N) = o(N)$, which is not implied. □

Note 19 (the identity is not accessible to computation). The finite-N residual $R(N, \theta') = \big|E_3(N;\theta') - (\tilde{r}(N) - \mathfrak{S}(N)(N - C(N)))\big|/N$ does not become small in any accessible range, and it cannot be made small by either free parameter. Over $N = 2 \cdot 10^5, 4 \cdot 10^5, 8 \cdot 10^5$ at $\theta' = 0.56$ it reads 0.4558, 0.3729, 0.3108, while the right-hand side it is compared against is $+0.01810311N$, $-0.01081673N$ and $-0.00633949N$ at those three N — of order $10^{-2}N$, falling in size, and changing sign between the first and the second, so that over the range that can be computed not even its sign is settled. Swept over $\theta' \in [0.51, 0.95]$, R is essentially monotone *increasing* — at $N = 8 \cdot 10^5$ from 0.170167 at $\theta' = 0.51$ to 4.991178 at $\theta' = 0.90$. The prediction registered before that sweep was run was an interior minimum, and the monotonicity refutes it; the mechanism offered for the monotonicity — that the finite-N error is dominated by the main-term cancellation over $m < M = N^{1-\theta'}$, which $\theta' \to 1$ destroys — is read off the sweep afterwards and is not itself tested. The best θ' is the smallest admissible one, just above $1/2$ — admissible in the asymptotic sense only, since at the sizes measured here the finite-N interval is empty (Note 9). Increasing N points the wrong way as well: at the best θ' the ratio of error to signal reads 15.19, 18.38, 26.84 at the three N above, because $C(N) = o(N)$ being true kills the right-hand side fast while the finite-N error dies only at a slow power of log. So Theorem 3 is a statement no computation here confirms, at any N swept and, on the trend of the residual, at any N a sweep of this kind would reach; the proof is the only access to it (`lab_theta_sweep.py`). The reach of that last clause is the reach of a trend read off the three values of N above, and it is offered as that.

Note 20. The mechanism is structural, not accidental. The identity $\mu * \log = \Lambda$ is the starting point of the Huang–Li argument (their (10)); it is therefore also what any divisor switch inside their $\log k$-weighted functional must return. No choice of θ', truncation, or smoothing can circumvent an identity. What the identity does not fix is the *strength* at which E_3 must be bounded, and that is Section 6.

6 How strong the surviving demand really is

Proof of Proposition 7. By Theorem 3, $\tilde{r}(N) = \mathfrak{S}(N)(N - C(N)) + E_3(\alpha) + O_A(N(\log N)^{-A})$. The triangle inequality gives $|C(N)| \le U(N) := \sum_{n<N} \Lambda(n)\mu^2(N-n)$, and $U(N) = \mathfrak{A}(N)N(1+o(1))$ — proved in [2, §3.3] itself by Bombieri–Vinogradov, with error $O(N(\log N)^{-A-1})$, and classically by Mirsky's theorem on squarefree shifted primes [4]. That theorem counts at the fixed sum and not uniformly in a cut, which is all that is needed here: the weight Λ is $(1 + o(1)) \log N$ on all but $\pi(N^{1-\varepsilon}) = o(N/\log N)$ of the terms, so no uniform partial summation enters. Hence $\mathfrak{S}(N)(N - C(N)) \ge \mathfrak{S}(N)(1 - \mathfrak{A}(N))N(1 + o(1))$, and $\mathfrak{A}(N) < 1$ for every N, so the right-hand side is positive. Substituting (7) leaves $\tilde{r}(N) \ge \varepsilon\, \mathfrak{S}(N)(1 - \mathfrak{A}(N))N(1 + o(1)) + O_A(N(\log N)^{-A})$. The threshold is $\gg N(\log N)^{-2}$ *uniformly* in N, which is what the quantifier of the statement needs: $\mathfrak{S}(N) \ge 2C_2 > 0$ for every N, since the factors over $p \mid N$ exceed 1; and if q_0 is the least prime not dividing N then $\prod_{p<q_0} p$ divides N, so $\theta(q_0) \le \log N + \log q_0$ and hence $q_0 \ll \log N$ by Chebyshev, whence

$$1 - \mathfrak{A}(N) \;\ge\; \frac{1}{q_0(q_0 - 1)} \;\gg\; \frac{1}{(\log N)^2},$$

the factor $1 - 1/(q_0(q_0 - 1))$ occurring in $\mathfrak{A}(N)$ and every other factor being at most 1. Note 21 gives the true order, $\asymp N/\log N$, on the family where the threshold is thinnest; the proof needs only some fixed power of $\log N$, because the two error terms are $O(N(\log N)^{-A-1})$ and $O_A(N(\log N)^{-A})$ with A at our disposal. Taking $A = 3$ makes both o of the threshold, so $\tilde{r}(N) > 0$. This is where the fixed ε is used: an unquantified $1 + o(1)$ would leave nothing to absorb them with.

For the size of the threshold, $\mathfrak{A}(N) = \prod_{q\nmid N}(1 - \frac{1}{q(q-1)})$ is largest when N carries the most small primes, so $1 - \mathfrak{A}(N)$ is small at primorials, where $1 - \mathfrak{A}(N) \asymp \sum_{q>p} \frac{1}{q(q-1)} \asymp 1/(p \log p)$ with $p \asymp \log N$. On that family $\mathfrak{S}(N)$ is not bounded but rises, and Note 21 carries the product. That the primorials are where the product is *least* is not proved here: what is shown is its order on that family, and what is measured is that the ten smallest margins over the even $N \le 1.6 \cdot 10^7$ are all of primorial shape (Measurement 22). □

Note 21 (the threshold's order, with the rise that accompanies the fall). Let N carry every odd prime up to y and no larger prime, so that $\log N = \theta(y)$. By Mertens' theorem

$$\log \frac{\mathfrak{S}(N)}{2C_2} = \sum_{3\le p\le y} \log\Big(1 + \frac{1}{p-2}\Big) = \sum_{p\le y} \frac{1}{p} + O(1) = \log\log y + O(1),$$

so $\mathfrak{S}(N) \asymp \log y$, while $1 - \mathfrak{A}(N) = \sum_{p>y} \frac{1}{p(p-1)}(1 + o(1)) \asymp 1/(y \log y)$ by partial summation against $\pi(t) \asymp t/\log t$. Since $\log N = \theta(y) \asymp y$ by Chebyshev,

$$\mathfrak{S}(N)\bigl(1 - \mathfrak{A}(N)\bigr) \asymp \frac{\log y}{y \log y} = \frac{1}{y} \asymp \frac{1}{\log N},$$

so the threshold on the adverse family is $\asymp N/\log N$. Discarding $\mathfrak{S}(N)$ as $\mathfrak{S}(N) \gg 1$ costs a factor $\log\log N$ at exactly the N where the threshold is hardest to meet, and since (7) asks E_3 to exceed the negative of this threshold, a sharper lower bound on it is a *weaker* hypothesis. The two factors move together: the same small primes that make $1 - \mathfrak{A}(N)$ fall make $\mathfrak{S}(N)$ rise.

Proof of Proposition 8. From Theorem 3, $|E_3(\alpha)| \le B_\mu(N)$, and $|C(N)| \le \mathfrak{A}(N)N(1+o(1))$ as above: $\tilde{r}(N) \ge \mathfrak{S}(N)(1-\mathfrak{A}(N))N(1+o(1)) - B_\mu(N) + O_A(N(\log N)^{-A})$. Substituting (9) leaves $\tilde{r}(N) \ge \varepsilon\,\mathfrak{S}(N)(1-\mathfrak{A}(N))N(1+o(1)) + O_A(N(\log N)^{-A})$, which is the right-hand side reached in the proof of Proposition 7; the uniform bound $\mathfrak{S}(N)(1 - \mathfrak{A}(N)) \gg (\log N)^{-2}$ established there dominates the error term at $A = 3$, so $\tilde{r}(N) > 0$. $\square$

Measurement 22 (the threshold, and the margin)**.** Over even $N \le 1.6 \cdot 10^7$ the threshold constant $\mathfrak{S}(N)(1 - \mathfrak{A}(N))$ has median 0.333459 and 0.1-percentile 0.119639; its minimum is 0.060890, attained at $N = 9\,699\,690 = 2 \cdot 3 \cdots 19$, the largest primorial in range, and the ten smallest values are all primorial-like. Multiplied by $\log N \log\log N$ the minimum over $N \ge 10^5$ is 2.482019, attained at $N = 510\,510$; over $N \ge 10^3$ it is 1.737799 at $N = 2310$; over $N \ge 16$ it is 0.810202 at $N = 30$. Every argmin is a primorial, which is the mechanism of Proposition 7 made visible (`lab_onesided_margin.py`).

Those three $\log N \log\log N$ figures are reported after the fact and were not the ones registered. That run registered the minimum of $\mathfrak{S}(N)(1 - \mathfrak{A}(N)) \log N \log\log N$ over the whole of its field, and the statistic is not defined at the bottom of that field: $\log\log N$ is negative for $N < e^e$, and the minimum came out -0.084557 at $N = 2$. The registered prediction was refuted; the field was then restricted, afterwards, to the one the rest of this paper measures on, and the three figures above are the restricted minima.

Two normalisations are in play and they do not agree, so both are printed. Along the four even primorials in $[10^3, 10^7]$ the product $\mathfrak{S}(N)(1 - \mathfrak{A}(N)) \log N$ runs 0.848929, 0.900116, 0.963553, 0.979580, a relative rise of 0.153900, while $\mathfrak{S}(N)(1 - \mathfrak{A}(N)) \log N \log\log N$ runs 1.737799, 2.100070, 2.482019, 2.721320, a relative rise of 0.565958 — more than three times as much. Over even $N \in [10^5, 1.6 \cdot 10^7]$ the infimum of $\mathfrak{S}(N)(1 - \mathfrak{A}(N)) \log N$ is 0.963553, at $N = 510\,510$. The factor $\log N$ is the normalisation this family obeys, and Note 21 says why: the extra $\log\log N$ over-corrects, because $\mathfrak{S}(N)$ rises along the same family (`audit_threshold_order.py`). The step that produces the margin is exact to five decimals: $U(N)/(\mathfrak{A}(N)N)$ has mean 0.999996 with standard deviation 0.000170 over the top octave. Checked against the truth, no even $N \in [10^5, 1.6 \cdot 10^7]$ has $\tilde{r}(N) < \mathfrak{S}(N)(1 - \mathfrak{A}(N))N$; the slack between them has minimum 3.7038, median 4.2902 and maximum 95.0889, the maximum at $N = 9\,699\,690$ — so the margin is thinnest exactly where the Goldbach count is largest (`lab_onesided_margin.py`).

Measurement 23 (what the weakening relocates)**.** At $\theta' = 0.56$ and $N = 2 \cdot 10^5$ through $3.2 \cdot 10^6$, the quantity $B_\mu(N)$ of (8) satisfies

$$B_\mu(N)/N = 0.8086,\ 0.7395,\ 0.7303,\ 0.6547,\ 0.5916$$

against a threshold of $0.3745N$, so the ratio $B_\mu(N)/(\mathfrak{S}(N)(1 - \mathfrak{A}(N))N)$ falls 2.159, 1.975, 1.950, 1.748, 1.580. The object Proposition 8 constrains is therefore already within a factor 1.6 of the threshold at the top of the accessible range, and falling; but a constant-factor bound valid for *all* large N is what is needed, and no computation supplies that. The ratio printed here is not the quantity that run registered in advance: it registered the saving of B over the Brun–Titchmarsh bound $N(\log N)^2$, and was refuted, because the measured B already carries those two powers of log. The figures above are the reading substituted after that refutation. That run registered four predictions and refuted three of them: the two Brun–Titchmarsh ones just described, and a third, that $|E_3|/B$ would fall below 0.05 — the signs buying a factor of twenty — where it measures 0.35 to 0.54. The fourth held, and it is the one the next sentence reads: (7) itself, whose margin the ratios above report, is not satisfied at the bottom of this range: at $N = 2 \cdot 10^5$ and $4 \cdot 10^5$ it fails, and holds from $8 \cdot 10^5$ on. That is not a counterexample to Proposition 7, which is a statement for all sufficiently large even N, but it is where the sweep sits relative to the condition it is measuring the margin of (`lab_onesided_demand.py`).

What $B_\mu(N)$ is small *because of* is not settled by any of this, and the control says so. Replacing $\mu(N-n)$ by a fair coin on the same support — the same Λ, the same K, the same weights, only the sign on $N-n$ changed — gives a *smaller* $B(N;K)$ at every N of the range, at 0.65459 to 0.73171 of the Möbius value, and by a margin larger than the spread across seeds. So the size of B measured here is a property of the support and the weights, and it is not evidence about μ. The cancellation across k is what the control does not touch — it leaves the signs $\mu(k)$ where they are — but that does not make the quantity which measures it, $|E_3|/B$, evidence about μ either, and the same run says so: on the coin that ratio is 0.0360 to 0.0961 against 0.3504 to 0.5413 for μ, so on this statistic the coin cancels five to ten times more. What the control establishes is negative in both directions. Neither the size of B nor the cancellation measured by $|E_3|/B$ separates μ from a sign pattern carrying no arithmetic at all, at the N reached here.

That control registered four predictions and two of them, C2 and C4, were refuted; the run records the reason as a defect of its own registration and does not repair it. The two bands had been copied from G1 and G2 of the demand run, which were already refuted there, so they were refuted here for the reason the originals were and they decide nothing about the control. What the control turns on is C1 — the coin's B is smaller at every N of the range — with C3 guarding it against the seed, and both hold (`lab_level_coin_null.py`).

Note 24 (the threshold is not a constant, and this sweep held it fixed). The 0.3745 above is $\mathfrak{S}(N)(1-\mathfrak{A}(N))$, and both factors depend on which primes divide N. The N of the sweep are $2 \cdot 10^5 \cdot 2^j$ with $2 \cdot 10^5 = 2^6 \cdot 5^5$, so all of them have odd radical 5 and *one* threshold. The five N measured above move the size of N by a factor 16, and the family swept in `audit_threshold_arithmetic.py`, which runs to $j = 7$, by a factor 128; neither moves the arithmetic at all.

Changing the arithmetic changes the verdict. At seven N in $[1.40 \cdot 10^6, 1.63 \cdot 10^6]$ chosen for their odd radicals the threshold runs from 0.073312 to 0.374487 — a factor of five, with the swept family sitting at the *maximum* rather than above the median 0.270682 — while $|E_3|/N$ moves the other way, from 0.1289 to 2.7763. The two compound: (7) holds at three of the seven and fails at four. The constraint is visible: when N carries several small odd primes, k must avoid them, and by the dilate identity of [9] so must m: at $N = 2 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17$ the smallest admissible k is 19, and the m below N/k are constrained to be coprime to $\mathrm{rad}(N)$. Why the inner sum then acquires a sign is not settled here. It is not that those m are almost all prime, so that $\mu(m) = -1$ dominates: the prime share among the contributing m falls steadily while the sign lock does not weaken, so the prime share is not what decides the sign; the parity of $\omega(m)$ is. The runs behind that reading are not in this packet, so no mechanism is asserted here — what is visible is the constraint, not its consequence. At $N = 2 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17$ all 513 terms of E_3 are negative and $|E_3| = B_\mu(N)$ exactly — no cancellation across k at all, against 442 positive terms and 562 negative ones at the swept N.

The same run registered a null and did not reject it. Asked whether the single constant 0.3745, used above and in place of each N's own threshold, would have reached a different verdict anywhere in the test set, it disagrees with the honest verdict at none of the seven:

the threshold moves by a factor of five while $|E_3|/N$ moves further and the other way, so the threshold is never the side that decides. What the arithmetic of N changes, it changes through E_3 and not through the constant.

The damage is not where it looks. Proposition 7 reaches its threshold through $|C(N)| \leq \mathfrak{A}(N)N$, and that bound is slack: $|C|/(\mathfrak{A}N)$ measures 0.0012 to 0.0208 across the seven. With the measured wall in its place, Theorem 3 asks for $|E_3|/N < \mathfrak{S}(N)(1 - |C(N)|/N)$, and *all seven* satisfy it (2.7763 against 5.3505 at the worst). The arithmetic dependence of the verdict therefore lives entirely in the slack of $|C| \leq \mathfrak{A}N$ and not in E_3. This is a measurement, not a proof — $C(N) = o(N)$ is the open problem and nothing here supplies it (`audit_threshold_arithmetic.py`).

Proposition 7 does not reopen the divisor switch. The no-go of [9] costs a factor $\exp(c_1\sqrt{(\frac{1}{2}+\delta)\log N})$, which exceeds $\log N \log\log N$ as it exceeds every power of $\log N$; so the weaker threshold is still out of reach of the weights that factor applies to, namely those whose transform Bombieri–Vinogradov reaches. It says nothing about the polynomials in $\log k$, which lie outside that hypothesis and are closed in [9] by the weaker argument recorded above.

7 The correction Δ to equation (18)

The published form of equation (18) drops a range. Several separate things follow and they carry different credit, and two of them are not ours.

The defect is not claimed as new: that the n-dependent restriction had been replaced by a fixed outer range was reported to the authors by S. Zheleznov before this note was written, and they have since prepared a corrected manuscript.

Neither is the repair from the inside, nor the demonstration that the omitted region is not empty. Moya [14], in August 2026, identifies the same missing moving cutoff, gives an exact finite counterexample showing that the displayed equation without it is not an identity, and shows that the published conditional theorem is recoverable under Huang–Li's own hypothesis, that hypothesis being uniform in the truncation variable and so controlling the moving endpoint directly. That note also formulates a moving-diagonal Möbius–logarithmic condition and reads the diagonal variant of [13] against the corrected rearrangement. All of that is prior to this section and is not repeated here.

What this section adds is the bound and what the bound is for. A correction that keeps the moving truncation repairs the derivation *from the inside*: the range $m \leq \alpha$ never appears, so there is no term to bound, and that is the route [14] takes. What follows is the *outside* version: the published form is taken as it stands, the discrepancy is named Δ, and it is bounded — a step [14] does not take, since on its route the term does not arise. The two are different kinds of work and neither replaces the other. The reason for taking the outside route here is not completeness: the term *is* the difference between the two formulations, and having it bounded is what lets a theorem proved against the fixed cut transfer to the corrected one (Proposition 25). Theorem 1 is proved against the published form, and without Δ it would say nothing about the corrected one.

The scope of the report has to be stated: we have worked from the arXiv version, and have checked the passage below in **both** v1 (May 2020) and v2 (August 2022), where it appears identically. We have not seen the version published in the Springer volume and make no claim about it.

Huang–Li define

$$S_2(\alpha) = \sum_{n<N} \Lambda(n)\,\mu^2(N-n)\,\tilde{\Lambda}_\alpha(N-n), \qquad \tilde{\Lambda}_\alpha(u) = \sum_{d|u,\ d>\alpha} \mu(d)\log(1/d),$$

and substitute $k = u/d$, so that the constraint $d > \alpha$ becomes

$$k \;<\; \frac{N-n}{\alpha}, \tag{24}$$

an n-*dependent* bound. Their displayed equation (18) reads

$$S_2(\alpha) \;=\; \sum_{k<\frac{N-1}{\alpha}} \mu(k) \sum_{\substack{n<N\\ n\equiv N\ (k)}} \Lambda(n)\mu(N-n)\log\Big(\frac{k}{N-n}\Big),$$

in which (24) has been replaced by the n-free bound $k < (N-1)/\alpha$. The two differ. Writing $N - n = mk$, the condition (24) is $m > \alpha$, so the right-hand side of (18) contains, in addition to $S_2(\alpha)$, exactly the terms with $m \le \alpha$. On those terms $\mu(k)\mu(N-n) = \mu(m)\mu^2(k)\mathbf{1}_{(k,m)=1}$ and $\log(k/(N-n)) = -\log m$, so

$$S_2(\alpha) \;=\; \text{RHS of (18)} \;+\; \Delta, \qquad \Delta \;=\; \sum_{2\le m\le\alpha} \mu(m)\log m \sum_{\substack{k<(N-1)/\alpha\\ (k,m)=1}} \mu^2(k)\,\Lambda(N-mk). \tag{25}$$

(The term $m = 1$ drops out since $\log 1 = 0$.) The trivial bound is $\Delta \ll N(\log N)^2$, which exceeds the target $O(N(\log N)^{-A})$, so Δ is not negligible and needs its own treatment.

It does close, by the machinery already used above and under hypotheses Huang–Li already assume. Indeed Δ has exactly the shape of the residual (16): the Möbius factor sits on the *short* variable $m \le \alpha$, and the long variable k carries only $\mu^2 \ge 0$. Since $\mu(m) \ne 0$ on the range of summation, the two conditions on k collapse into one,

$$\mu^2(k)\,\mathbf{1}_{(k,m)=1} \;=\; \mu^2(mk), \tag{26}$$

and expanding $\mu^2(mk) = \sum_{b^2|mk} \mu(b)$ with the truncation $b \le B_1 := (\log N)^{A+4}$ — the same one Huang–Li use in their §3.1, where it is called B; the subscript keeps it apart from the free exponent of Lemma 13 — reduces Δ to sums of Λ over progressions to moduli $[m, b^2] \le \alpha B_1^2 = \alpha(\log N)^{2A+8}$. The tail $b > B_1$ is discarded as the d-tail of Step 4 was, but counted on the pair (m, k) and not on the modulus alone. Such terms have $b^2 \mid mk$, which holds exactly when $k \equiv 0$ to the modulus $b^2/(b^2, m)$; so for each m the number of admissible $k < K := (N-1)/\alpha$ is at most $K(b^2, m)/b^2$, and

$$\sum_{m\le\alpha} (b^2, m) \;=\; \sum_{g|b^2} g\cdot\#\{m\le\alpha : (b^2,m) = g\} \;\le\; \sum_{g|b^2} g\cdot\frac{\alpha}{g} \;=\; \alpha\, d(b^2).$$

Hence, with $\Lambda \ll \log N$ and $K\alpha = N-1$,

$$\ll \log N \sum_{b>B_1} \frac{K\alpha\, d(b^2)}{b^2} \;\ll\; N\log N \sum_{b>B_1} \frac{d(b^2)}{b^2} \;\ll\; \frac{N\log N\,(\log B_1)^2}{B_1} \;\ll\; \frac{N(\log\log N)^2}{(\log N)^{A+3}},$$

which is $\ll_A N(\log N)^{-A}$. The last step is $\sum_{b>X} d(b^2)b^{-2} \ll (\log X)^2/X$, by partial summation from $\sum_{b\le Y} d(b^2) \ll Y(\log Y)^2$.

Counting on the pair is what makes this step unconditional in θ', and the distinction is not cosmetic. Bounding each progression separately by $N/[m, b^2] + 1$ with $[m, b^2] \ge b^2$ drops the factor $1/m$ and drops the constraint that an admissible k exists at all; the resulting $+1$ then contributes $\alpha\sqrt{N}\log N$, which is a count of pairs and carries no level of distribution, so it would be admissible only for $\alpha < \sqrt{N}$ and no improvement in the level would move it. The bound above has no such term. What is left is a main term $\mathfrak{A}(N)\sum_{m\le\alpha} \mu(m)\lambda(m)(\log m)\,\mathcal{T}_m/m$ which is $O(Ne^{-c\sqrt{\log N}})$ by Lemma 16 and partial summation.

The density here is not the one Lemma 15 proves. That lemma evaluates the first of

$$\sum_{d\geq 1}\sum_{e|m}\frac{\mu(d)\mu(e)\mathbf{1}_{(d,N)=1}}{\varphi(m\,\mathrm{lcm}(d^2,e))},\qquad \sum_{b\geq 1}\frac{\mu(b)}{\varphi([m,b^2])},$$

and what the expansion above leaves is the second, over the moduli restricted to $(\,\cdot\,,N)=1$. The two agree, and the check is the same local computation: for squarefree m the factor of the second is $\frac{1}{p-1}-\frac{1}{p(p-1)}=\frac{1}{p}$ at $p\mid m$ and $1-\frac{1}{p(p-1)}$ at $p\nmid m$, and the restriction sets the latter to 1 at $p\mid N$ exactly as in Lemma 15 — the point Note 12 makes, here in its second use. The product is $\mathfrak{A}(N)\lambda(m)/m$, as written.

The level is then exactly the one their hypothesis supplies, with no enlargement of A. Hence:

- In the Corollary-1 regime, $\alpha=N^{1-\theta'}<N^{1/2}$ and Bombieri–Vinogradov closes Δ *unconditionally.*
- In the general Theorem-1 regime, $\alpha\asymp N^{\theta}$ and Δ closes under the hypothesis $EH(N^{\theta}(\log N)^{2A+8})$, which is assumed there.

So [2]'s Theorem 1 and their Corollary 1 stand as stated. In the published text the treatment of Δ is missing; in the corrected text Δ does not arise, because the truncation is not moved off the inner variable in the first place.

7.1 The two formulations differ by Δ

The corrected argument keeps the moving cut: with $Y_k=\lceil N-\alpha k\rceil-1$ the inner sum stops at Y_k, the terms with $m\leq\alpha$ never enter, and (25) has nothing to correct. Every statement above was proved against the fixed cut, so the question is whether it survives the move. It does, and the term that carries the difference is Δ itself.

Write $J=T_1(N-1)$ for the fixed-cut object at its extreme endpoint and $T_1^*=\sum_{k<K,(k,N)=1}\mu(k)E_\mu(Y_k;k)$ for the moving-cut one. The difference is carried by the n the moving cut drops, and there

$$n>Y_k\iff N-n\leq\alpha k\iff m\leq\alpha,\qquad m=\frac{N-n}{k},$$

so on exactly those terms $\mu(k)\mu(N-n)=\mu(k)\mu(mk)$, which vanishes unless mk is squarefree and equals $\mu(m)\mu^2(k)$ otherwise: the Möbius factor on the *short* variable, the long one carrying only $\mu^2\geq 0$. That is (25)'s shape with weight 1 in place of $\log m$.

Proposition 25 (the bound survives the corrected cut)**.** *Fix $\theta'\in(1/2,1)$, put $\alpha=N^{1-\theta'}$ and $K=(N-1)/\alpha$. Then for every $A>0$, $T_1^*\ll_{A,\theta'}N(\log N)^{-A}$, unconditionally.*

Proof. Exchanging as above, $J-T_1^*=\Xi_{\text{main}}-\Xi_{\text{mean}}$ with

$$\Xi_{\text{main}}=\sum_{m\leq\alpha}\mu(m)\sum_{\substack{k<K\\(k,mN)=1}}\mu^2(k)\,\Lambda(N-mk),\qquad \Xi_{\text{mean}}=\sum_{\substack{k<K\\(k,N)=1}}\frac{\mu(k)}{\varphi(k)}\big(C(N-1)-C(Y_k)\big),$$

the constraint $n\geq 1$ being automatic since $m\leq\alpha$ and $k<K$ force $mk<N-1$.

$J\ll_{A,\theta'}N(\log N)^{-A}$ is Theorem 1 at $t=N-1$, which lies in the range of its supremum, and it is used exactly where Theorem 1 needs it. It is one of the two places this proof needs $\theta'>1/2$; the other is the bridge term below, whose moduli carry the same inequality (Note 9).

In Ξ_{main} the term $m=1$ is $\sum_{k<K,(k,N)=1}\mu^2(k)\Lambda(N-k)\leq\sum_{N-K<j<N}\Lambda(j)\ll K\log N=N^{\theta'}\log N$, which is $\ll_A N(\log N)^{-A}$ because $\theta'<1$. The terms $2\leq m\leq\alpha$ are (25) with $\log m$ replaced by 1 and close by the argument above, the main term dying by Lemma 16 applied directly — the weight being 1, no partial summation is needed to remove a $\log m$. One step is not verbatim: there the inner variable carries $(k, m) = 1$ and (26) absorbs it into $\mu^2(mk)$, while here it carries $(k, mN) = 1$, and the $(k, N) = 1$ that is left over after (26) is discarded by Lemma 11 at a cost of $O(N^{o(1)})$, as at (16).

For Ξ_{mean}, since $n > Y_k$ iff $k > R_n := \lceil (N-n)/\alpha \rceil - 1$, exchanging gives $\Xi_{\text{mean}} = \sum_{n<N} a_n(\rho_N(K-1) - \rho_N(R_n))$, where

$$\rho_N(R) := \sum_{\substack{d \le R \\ (d,N)=1}} \frac{\mu(d)}{\varphi(d)} \ll e^{-c_1\sqrt{\log R}} \qquad (\log N \ll \log R)$$

by [2, Lemma 1], a form of [1, Lemma 2.1] — the hypothesis $\log N \ll \log R$ is part of the statement and is checked below — and $a_n = \Lambda(n)\mu(N-n)$. Split at $N - n = H$ with $H := 2N^{1-\theta'/2}$. On $N - n \le H$ use $|\rho_N| \ll 1$ and $\sum_{N-H \le n < N} |a_n| \ll H \log N$, which is $\ll_A N(\log N)^{-A}$ since $1 - \theta'/2 < 1$. On $N - n > H$ one has $R_n \ge N^{\theta'/2}$, so both arguments of ρ_N meet its hypothesis against $n = N$ and both terms are $\ll e^{-c\sqrt{\theta' \log N}}$; with $\sum_{n<N} |a_n| \ll N$ this is $\ll_A N(\log N)^{-A}$ as well. □

Note 26 (the two cut-offs, and the gap between them). The letter K carries two values in this paper. Huang–Li's functionals are defined with $K = (N-1)/\alpha$, while Theorem 1 is stated with $K = N^{\theta'}$, and the three places that apply the theorem to those functionals — Corollary 2, Steps 0–1 of Section 5, and the fixed-cut term of this proposition — use the first where the theorem supplies the second. In the Corollary-1 regime $\alpha = N^{1-\theta'}$, so the two differ by $(N-1)/\alpha = N^{\theta'} - N^{\theta'-1}$: an interval of length below 1, containing at most one integer k. That term is $|\mu(k)E_\mu(t;k)| \ll N^{1-\theta'} \log N$, which is $\ll_A N(\log N)^{-A}$ since $\theta' > 0$, so the substitution is free. It is written out here because the paper otherwise defines the same letter twice without saying so.

Note 27 (what the proposition does not cover). It is stated in the Corollary-1 regime only. In the general Theorem-1 regime $\alpha \asymp N^{\theta}$ the outer truncation is $K \asymp N^{1-\theta}$, which drops below $N^{1/2}$ as soon as $\theta > 1/2$, and Theorem 1 is stated for $K = N^{\theta'}$ with $\theta' \in (1/2, 1)$. The exchange and the estimate of Ξ_{main} are unaffected — the latter closes under the *EH* assumed there, as Δ does — but the fixed-cut input is not available, so no claim is made outside $\theta' > 1/2$. The no-go of [9] is stated on the same threshold and for the same reason — each needs the cofactor $N^{1-\theta'}$ to be a level at which Λ is known to distribute — but the obstruction it exhibits is a different object from this threshold, and Note 9 separates the two.

The exchange, the split of Ξ and the shortness of its variable were verified as identities rather than as estimates, over $N = 2 \cdot 10^5$ to $3.2 \cdot 10^6$ at $\alpha = N^{0.44}$: the two sides of each agree, at worst over that range, to $1.478 \cdot 10^{-17}$ and $4.547 \cdot 10^{-18}$ relative to N — five orders below the tolerance registered for the run — accumulated in index orders that share nothing, and the largest m carrying a surviving term is 213, 291, 393, 533, 727 against α = 215, 292, 396, 537, 728 (`audit_delta_bridge.py`). Those five are the largest m inside the sum as it is enumerated, which stops at $\lfloor \alpha \rfloor$. The run's rule Z3 asked whether any surviving m exceeds α, and until 2026-09-05 it compared that enumeration against its own bound and could not fail. Carrying the same survival test past α refutes it: surviving terms are found throughout $m \le 4\alpha$, so the bridge tail is not confined to the short variable. The two identities above are unaffected — they are Z1 and Z2, and both hold.

8 Relation to the literature

This section is a placement, and it is deliberately conservative about what is offered here as new.

The hypothesis is Murty–Vatwani's. [5] posed the shifted Möbius Elliott–Halberstam conjecture EH_{μ_h} and proved that, with EH_Λ, it makes $\sum_{n\le x}\Lambda(n)\Lambda(n+h)\sim\mathfrak{S}(h)x$ and $\sum_{n\le x}\Lambda(n)\mu(n+h)=o(x)$ equivalent, and that the twin prime conjecture follows. [2] transposed the hypothesis and the argument to the Goldbach shift; [6] develops the equidistribution side. Everything from Section 2 onward is written against [2]'s formulation, because that is the one stated for $N-n$, but the decomposition, the μ^2 restriction and the split at y are [5]'s. On the Goldbach side the precursor is Pan [11], which this paper has not been able to consult and does not rely on; what [2] record of it is that Pan truncates at $Q=N^{1/2}(\log N)^{-20}$, below the square root, and leaves $R_3(N)$ untouched without giving any hint how to approach it, while estimating $R_3(N)$ through the equidistribution of $\Lambda(n)\mu(N-n)$ is what [2] call the main thrust of their memoir.

The mechanism is standard. A divisor switch that moves the work onto a short variable, followed by Bombieri–Vinogradov, is ordinary practice; the completion of the divisor sum is the step that makes the cofactor short, and Section 2 says why it is not a formality. Every ingredient (Bombieri–Vinogradov [8]; the Goldston–Yıldırım estimate [1] as used in [2], carrying its hypothesis $\log n\ll\log x$; the elementary density identity of Lemma 15; the classical zero-free region estimate for $\sum\mu(a)/a$ of Lemma 16) is off the shelf. Theorem 1 is offered as a *lemma about the Huang–Li reduction*, not as a contribution to the theory of Bombieri–Vinogradov, and what difficulty it has lies in the bookkeeping — in particular that the condition $(k,N)=1$ must be discarded by the degeneracy argument (Lemma 11) rather than expanded over $e\mid N$, and that main terms may be assigned only to classes with $(q,N)=1$, the second of which, if missed, multiplies the density of the whole computation by $\prod_{p\mid N}\big(1-\frac{1}{p(p-1)}\big)$ — a factor below 1 at every even N (Note 12).

Theorem 3. We are not aware of the identity in the literature. Its ingredients are again classical; what it does is locate, exactly, where the Goldbach content of the Huang–Li reduction sits — in the weight $\log k$, by $\mu*\log=\Lambda$.

The companion no-go. That E_3 cannot be reached from E_4 by reweighting — that no weight between $w_k=1$ and $w_k=\log k$ whose transform $\mu*w$ is supported where Bombieri–Vinogradov reaches extracts $C(N)$ by divisor switching, even granting $EH(N^{\theta_E})$ for every fixed $\theta_E<1$ — is proved in [9] and is not reproduced here. The weights outside that support hypothesis, the polynomials in $\log k$, are treated there separately and by a weaker argument. In genre it is a precise form, for this reduction, of the phenomenon Bombieri's asymptotic sieve [21] records, and a reader who finds it unsurprising is right to.

What Theorem 1 adds, and how far the search behind that claim reaches. Both papers assume their hypothesis on the flat branch. Murty and Vatwani [5] deduce equidistribution for $\Lambda(n)\mu(n+h)\log(n+h)$ from EH_{μ_h} by partial summation (their Proposition 4.2) and bound S_4 with it (their Proposition 4.3, stated under "$x/y\le x^\eta$, and $EH_{\mu_h}(x^\eta)$ holds"); Huang and Li [2] use their Lemma 5, stated under "Suppose the conjecture $EH_\mu(N^{1-\theta})$ holds", at the same place; and Cantarini [13], who replaces EH_μ by a diagonal hypothesis in the fixed class $n\equiv N\ (q)$ with $(N,q)=1$, still requires it on both branches — "the same strength of cancellation both in the pure Möbius case and in the logarithmically weighted case". Theorem 1 is the assertion that the flat branch carries no hypothesis at all.

That assertion is to be kept apart from the idea of the reduction itself, which is older than [2] and is in print. Reviewing [11] in 1983, Shiu [12] observes that the number of terms in Pan's $R_3(N)$ decreases as the truncation Q increases, and is "led to speculate if the Goldbach conjecture could be derived from an eagerly awaited extension of Bombieri's theorem in the direction of H. Halberstam's conjecture". That is the shape of [2]'s Corollary 1, stated forty years earlier, and nothing below is a claim to it. It is also not a statement about the object of Theorem 1: it is a speculation, it concerns the $\Lambda\cdot\Lambda$ correlation and an extension of a theorem about primes, it carries no Möbius twist and no signed sum, and its direction is the opposite one — it asks for a stronger input, where Theorem 1 removes an input at one of the two places that consume it.

The search for a prior statement of it reaches as follows, and its boundary is given so that it can be checked. The forward citation graphs of [5], [6], [2] and [16] were read in full in Semantic Scholar and OpenAlex in August 2026 — 22 and 13 records respectively for [5], 2 for [6] (both of them [2]), 2 for [2] (namely [13] and the survey [15]), and 8 for [16]; the reference lists of [5] (27 items), [2] (18), [6] (23) and [13] (28) were read one level out; and zbMATH Open together with the arXiv metadata were queried on "Elliott–Halberstam" with "Möbius", "twisted Elliott–Halberstam", "Möbius function" with "shifted primes", "level of distribution" with "Möbius function", "Möbius" with "arithmetic progressions to large moduli", "Bombieri–Vinogradov theorem" with "Möbius function", "Chowla conjecture in arithmetic progressions", "von Mangoldt" with "Möbius" and "correlation", "Goldbach" with "Elliott–Halberstam", and "divisor switching", and — since the vocabulary of the older literature is not that of EH_μ — also on "Halberstam conjecture" with "Goldbach" and "Bombieri", on "extension of Bombieri's theorem" with "Halberstam's conjecture", and on [11]'s title. The last three reach [12] and [11], together with Goldston–Pintz–Yıldırım's *Primes in tuples I* and the Polymath equidistribution estimates of Zhang type, both of which concern the level of distribution of the primes and not a μ-twisted correlation. Inside that set the papers stating a hypothesis of EH_μ type are [5], [6], [2], [13] and the survey [15]; each states the flat branch as a hypothesis or as a conjecture, none states it as a theorem, and no unconditional instance of $EH_\mu(N^\theta)$, at any level $\theta > 0$, appears in any of them.

The unconditional statements nearest to Theorem 1 inside that set differ from it in a way that can be named in each case. A Bombieri–Vinogradov theorem for μ itself is classical ([19]; Motohashi's induction principle [20]), but its object is μ in progressions and not a correlation, and the induction principle applies to convolutions, which $\Lambda(n)\mu(N-n)$ is not. [6] proves unconditionally that a Bombieri–Vinogradov theorem for f extends to $f(n)\mu^2(n+h_1)\cdots\mu^2(n+h_k)$ at almost the same level, and [5] evaluate $\sum_{n\le x}\Lambda(n)\mu^2(n+h) \sim A_h x$ by Bombieri–Vinogradov; in both the twist is μ^2, which carries no sign, and the level is the level of f. [16] and [17] obtain cancellation in $\sum_{p\le X}\mu(p+h)$ and in Hardy–Littlewood–Chowla correlations for almost all shifts $h \le H$, and [18] gives an asymptotic for those correlations on the hypothesis that Siegel zeros exist; all three concern the mean value that Theorem 3 isolates as $C(N)$, and not the distribution of $\Lambda(n)\mu(N-n)$ in progressions. Against each of them the same three features of Theorem 1 do the separating: the sum over moduli is signed by $\mu(k)$ rather than taken in absolute value, the conclusion is unconditional, and what it leaves is a named residual rather than a hypothesis.

The fixed residue class is deliberately *not* on that list. It is a convenience the literature already grants: [5] and [2] both record that their arguments go through in a fixed-class form, and [8, Notes to §20.5] treats releasing the maximum over residue classes as a known route to a higher level rather than as a novelty. The class being fixed makes Theorem 1 easier to prove; it is not a respect in which its statement is new.

What the search does not cover is stated with the same explicitness. [11] itself has not been consulted; only [12], which is a review of it and says so. [6] has been read through its published abstract and its reference list and not in the original; MathSciNet was not available and zbMATH Open was used in its place. The arXiv carries no full-text index, so a statement made in the body of a paper whose title and abstract do not mention μ-twisted equidistribution would not have been reached. The forward sweep was not carried past one level for the classical items in those reference lists, and no systematic search was made of the non-English literature beyond what the lists themselves carry. The greater part of the query set is phrased in the vocabulary of EH_μ, and [12] shows what that costs: a statement of the same shape can sit in the older literature under "Halberstam's conjecture" and "an extension of Bombieri's theorem" and be missed. The three queries added for that vocabulary recovered [12]; whether the same gap hides anything else is not settled here.

Beyond the square-root barrier. Lichtman [3] obtains a level of distribution past $1/2$ for the primes, which postdates [2]. Note 9 and [10] place Theorem 1 against it: the $1/2$ in Theorem 1 is the level of distribution and nothing else; [3]'s $153/256$ is uniform over $0 < |a| < x^{1+\varepsilon}$ and so covers the class $n \equiv N$; and what stops it applying here is that the coefficient of (18) is not well-factorable at that level. Whether an analogue for μ on shifted primes would move the regime of Theorem 3 is a different question and remains open.

Behind that there is a question about the weight rather than about the level, and it is well posed here because Theorem 3 identifies which weight carries the Goldbach content. Since [2] require $EH_\mu(N^{\theta'})$ for a single $\theta' > 1/2$, and since every level past $1/2$ now on record is stated for well-factorable or triply well-factorable coefficients, one may ask whether the weight $w_k = \log k$ that the consumption actually requires lies in that class. It does not, at any level, and the reason is short enough to give here rather than to cite. Write $\lambda_k = \mu(k)(\log k/\log K)\mathbf{1}_{k<K,\ (k,N)=1}$ for the normalised weight the consumption requires. At a prime $p < K$ with $p \nmid N$ one has $\lambda_p = -\log p/\log K \neq 0$, so λ is supported on primes all the way up to $K = N^{\theta'}$. A triply well-factorable sequence of level N^{ϑ} is by definition a combination of terms each factoring into three pieces of level at most $N^{\vartheta/3}$, so every such sequence, and every finite combination of them, vanishes at moduli above $N^{\vartheta/3}$. For $\theta' > 1/2$ and any $\vartheta \le 1$ we have $\vartheta/3 < 1/2 < \theta'$, so the primes p with $N^{\vartheta/3} < p < N^{\theta'}$ lie in the support of λ and outside that of every such combination. Hence λ is not one of them.

Nothing in this concerns the truth of any level-of-distribution statement for μ — what is denied is membership of the class such a statement would be about. A fuller treatment is in [10], on which the paragraph above does not depend.

9 Summary

- Theorem 1: the Möbius-weighted, fixed-class correlation sum with weight $w_k = 1$ is $\ll_A N(\log N)^{-A}$, unconditionally, for any level $N^{\theta'}$ with $\theta' \in (1/2, 1)$.
- Corollary 2: hence Huang–Li's E_4 (their Lemma 5) needs no hypothesis, and the EH_μ demand of their §3 collapses to the scalar E_3.
- Theorem 3: that scalar is unconditionally equivalent to Huang–Li's equation (22), hence already gives binary Goldbach for large even N, and gives the asymptotic $\tilde{r}(N) \sim \mathfrak{S}(N)N$ exactly when $C(N) = o(N)$.
- Propositions 7 and 8: Goldbach itself needs only a one-sided bound on E_3 at a threshold $\asymp N$, and a constant-factor bound where EH_μ is consumed with a saving of $(\log N)^A$ for every A; the two are not bounds on the same object, and the implication runs one way.
- Proposition 25: the bound of Theorem 1 holds against the corrected formulation of [2], unconditionally for $\theta' \in (1/2, 1)$. The two formulations differ by a term of Δ's shape, so the fixed cut is not superseded — it is one of the two ingredients and the omitted term is the other.
- Section 7: equation (18) of [2] omits an n-dependent constraint; the missing term Δ is exhibited and closes under hypotheses already assumed.
- Section 1.4: the net progress toward Goldbach is zero.

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Disclosure statement

No potential competing interest, financial or non-financial, was reported by the author. No funding was received for this work. An AI assistant was used in preparing it; what tool, how, and why is set out in Appendix B's preceding section, together with what was checked by computation and what was not checked at all.

Data availability statement

The code behind every figure printed here, the output it produced, and the Lean 4 development are openly available at `https://github.com/soke91/arithmetic-convolution-fields`, in the directory `P1-mobius-fixed-class/`. One class of figure has no script and says so in Appendix B: the citation counts and the exponents quoted from other papers, which are read from those papers and from the databases named there. No data derived from human or animal subjects were used; every quantity here is computed from the integers.

A Use of an AI assistant, and what checked what

This work was done in close collaboration with an AI assistant, which carried much of the derivation and all of the computation. The author takes full responsibility for every claim made here.

Which tool, and how. The manuscript in its present form was prepared with Anthropic's Claude, model Opus 5, run through the Claude Code command-line interface. It was used to derive and check the identities, to write and run every script in the accompanying code, and to draft and revise this text. Earlier stages used earlier models of the same family; the repository does not record which, and nothing is relied on from those stages that has not since been rederived and rerun.

What follows says what was checked and by what, rather than asserting that it was checked.

Machine-verified. The finite rearrangements the proof of Theorem 1 runs on are formalised in Lean 4 against Mathlib, in the file `lean/GoldbachLean/Fixedclass.lean` of the code packet: that the subtracted mean term of (11) leaves the k-sum, the exchange and the reindexing that together make the divisor switch (13), the split of σ_K into a complete sum minus a tail that gives (14), and the factorisation $u = mk$ of (16). Each depends on `propext`, `Classical.choice` and `Quot.sound` and on nothing else; the axiom report is in the packet. The statements take Λ, μ, φ and the weight as arbitrary functions, so what is verified is that these steps use no arithmetic property of them — which is what licenses the claim that E_3 and E_4 differ in one factor and nothing else.

Two of the formalised statements are there to be contradicted rather than used. One exhibits a subtracted term depending on k for which (11) fails, so that the step is seen to rest on the k-independence of $C(t)$ and not on algebra alone; the other exhibits index sets carried onto each other by $n \mapsto N - n$ for which (13) still fails, so that the congruence-to-divisibility translation is seen to carry the step rather than the reindexing.

Theorem 1 itself is not machine-verified, and no statement in this paper has had a reading by a subject expert. It rests on Bombieri–Vinogradov, which Mathlib does not carry, so no part of it is formalised; (10), Lemma 10 and Lemma 11 are arithmetic and are not formalised either. The proofs are as given in the text.

Checked by computation. Every figure quoted is produced by a named run in the accompanying code, with the one exception recorded in Appendix B: the citation counts and the exponents quoted from the literature in §8, which are read from the cited papers and named databases on the date given there. A check in the build refuses a printed figure that no cited run produces. Every registered rule is reported with its verdict, including the thirteen the runs did not meet.

Cross-read. The proofs and the computations were additionally read by separate AI sessions working without sight of each other's findings, and several statements here are corrections those readings produced. That is a check on internal consistency, on arithmetic, and on whether each statement matches the run it cites. It is not a substitute for either of the two checks named above as absent.

B Reproducibility

The code and the output it produced are at

`https://github.com/soke91/arithmetic-convolution-fields`

in the directory `P1-mobius-fixed-class/`. Each entry in the table below is run as `python code/<script>.py` from that directory, and each writes the file of the same name in `results/`. The figures printed here were produced under Python 3.12.10 and NumPy 2.2.5. That address is mutable, so a number checked against the branch head may have been recomputed since; the release named there pins the state these figures were read from.

A nonzero exit status is not a failure. These runs carry preregistered rules, and a run whose own rule was refuted says so in its exit code as well as in its output. Forty-two rules carry a verdict across the twelve runs and thirteen of them were refuted; all are reported below.

Every numerical figure printed above comes from one of the following scripts, with one class of exception: the citation counts, the reference-list lengths and the exponents quoted from cited papers in Section 8 are read from the literature and from the two databases named there on the date named there, and no script produces them. Each script below is standalone, runs under Python with numpy on a laptop, prints its own pass/fail criteria, and writes one result file. Seven of the twelve runs record a prediction their own output refuted, and all seven are reported: Notes 19, 9 and 12, Measurement 22, Measurement 23, which carries two —

the demand run and its control — and §7.1, where the rule Z3 of `audit_delta_bridge.py` is recorded as refuted.

Script	Result file	Supports
`audit_switch_identity.py`	`audit_switch_identity.txt`	(13)
`audit_delta_bridge.py`	`audit_delta_bridge.txt`	§7.1
`audit_threshold_arithmetic.py`	`audit_threshold_arithmetic.txt`	Note 24
`audit_density_identity.py`	`audit_density_identity.txt`	Lemma 15
`audit_E3_constant.py`	`audit_E3_constant.txt`	(23)
`lab_theta_sweep.py`	`lab_theta_sweep.txt`	Note 19
`lab_onesided_margin.py`	`lab_onesided_margin.txt`	Measurement 22
`audit_threshold_order.py`	`audit_threshold_order.txt`	Measurement 22, Note 21
`lab_onesided_demand.py`	`lab_onesided_demand.txt`	Measurement 23
`lab_level_coin_null.py`	`lab_level_coin_null.txt`	Measurement 23
`audit_theta_halfline.py`	`audit_theta_halfline.txt`	Note 9
`audit_trap_density.py`	`audit_trap_density.txt`	Note 12

No computation is used as a premise of any proof. The computations verify exact identities ((13), Lemma 15), fix the sign of a constant that the analysis determines ((23)), or measure a quantity whose size is stated only over the range measured.